Weather Regressiom

Grishma Gajurel

2025-11-08

```
# Load required packages
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(tidyr)
library(lubridate)

## Warning: package 'lubridate' was built under R version 4.3.3

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union

library(purrr)

## Warning: package 'purrr' was built under R version 4.3.3

library(broom)

## Warning: package 'broom' was built under R version 4.3.3

library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.3.3

# Load all csv's
weather_2022 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Abiotic Data all years .xlsx - 2022.csv")%>%
  select(-1) # Not considering the first row/column since its all NA
weather_2023 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Abiotic Data all years .xlsx - 2023.csv")%>%
  select(-1)
weather_2024 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Abiotic Data all years .xlsx - 2024.csv")%>%
```

```r
  select(-1)
weather_file <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Weather Data - Sheet1.csv")
weatherdb_2022 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Weather Database - 2022.csv")
weatherdb_2023 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Weather Database - 2023.csv")
weatherdb_2024 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Weather Database - 2024.csv")
weatherdb_2025 <- read.csv("~/Documents/Ecology /USDA Grant/weather
data/Weather Database - 2025.csv")

# Convert Date column to uniform DD-MM-YYYY format for weather_file and
selecting required column
weather_file <- weather_file %>%
  mutate(
    Date = parse_date_time(Date, orders = c("d/m/Y", "d-b-Y", "m/d/Y")),
    Year = year(Date),
    Date = format(Date, "%d-%m-%Y"),

    # Temperature: Celsius -> Fahrenheit
    Temperature..degree.fahrenheit.. = ifelse(
      !is.na(Temperature..degree.celsius..),
      (Temperature..degree.celsius.. * 9/5) + 32,
      NA
    ),

    # Wind speed conversions into mph
    Avg..Wind.Speed..mph. = case_when(
      !is.na(Average.wind.speed..m.s.) ~ Average.wind.speed..m.s. * 2.23694,
# m/s → mph
      !is.na(Avg..Wind.Speed..km.h.) ~ Avg..Wind.Speed..km.h. / 1.60934,    #
km/h → mph
      TRUE ~ NA_real_
    ),

    # Convert text time to POSIXct (meaning changing 9:47 AM into TIME format
HH:MM:SS which is 09:47:00, its a function name dw abt it)
    Time_24 = format(parse_date_time(Time, orders = c("h:M p", "H:M")),
"%H%M"),
    # Convert to numeric
    Time_24 = as.numeric(Time_24),
    # Extract last two digits (minutes)
    minutes = Time_24 %% 100,
    # Custom rounding rule: >30 round up, else round down
    TimeRounded = ifelse(minutes > 30,
                         ceiling(Time_24 / 100) * 100,    # round up
                         floor(Time_24 / 100) * 100)      # round down
  ) %>%
```

```r
  select(
    Location, Year, Date, TimeRounded, everything(),
    -any_of(c(
      "Notes", "Collector.Name", "Team", "Weather.condition", # you can
remove Time_24 and minutes to see whats really happening with POSIXct
      "X..Trees.Blooming", "Temperature..degree.celsius..",   # you can also
remove some of them from this quotations as needed, just removed to maintain
clean data
      "Average.wind.speed..m.s.", "Avg..Wind.Speed..km.h.",
      "minutes", "Time_24", "Cloud.cover"
    ))
  ) %>%
  rename(
    `Temperature (F)` = Temperature..degree.fahrenheit..,
    `Humidity (%)` = Relative.humidity..,
    Pressure = Barometric.Pressure..mmHg,
    `Hour (CST)` = TimeRounded,
   `WindSpeed (mph)` = Avg..Wind.Speed..mph.
  )

# Doing the same as above but for the weatherdb
all_weatherdb <- bind_rows(weatherdb_2022, weatherdb_2023, weatherdb_2024,
weatherdb_2025)

all_weatherdb <- all_weatherdb %>%
  mutate(
    Date = format(make_date(YEAR, MONTH, DAY), "%d-%m-%Y")
  ) %>%
  select(Date, YEAR, everything(), -MONTH, -DAY) %>%
  rename(
    Year = YEAR,
    `Hour (CST)` = HOUR.CST,
    `Temperature (F)` = AVG.TEMP.F,
    `WindSpeed (mph)` = WIND.SPEED.MPH,
    `Humidity (%)` = AVG.REL.HUMIDITY..,
    Pressure = PRESSURE.INCHES
  )

# ignoring the blank row 14 (the row above Note) and changing the date for
2024 datasheet
weather_2023 <- weather_2023[-14, ]
weather_2024[1, ] <- gsub("27-Apr-2024", "27-Apr-24", weather_2024[1, ])

# Function to pivot wide into long for weather_2022, weather_2023, and
weather_2024
process_weather <- function(weather_data) {
  # Removing note row from the file 2023
  weather_data <- weather_data[!grepl("^\\s*Note", weather_data[[1]],
ignore.case = TRUE), ]
```

```
# Extract headers
dates_row <- weather_data[1, ] %>% unlist() %>% as.character()
weeks_row <- weather_data[2, ] %>% unlist() %>% as.character()
types_row <- weather_data[3, ] %>% unlist() %>% as.character()

# Extract data starting from row 4
data <- weather_data[4:nrow(weather_data), ]

# Build column names
col_names <- "Location"
week_counter <- 1
week_dates <- c()

for (i in 2:ncol(data)) {
  if (!is.na(dates_row[i]) && dates_row[i] != "N/A") {
    current_date <- dates_row[i]
  }

  if (!is.na(types_row[i]) && grepl("Temperature", types_row[i],
ignore.case = TRUE)) {
    col_names <- c(col_names, paste0("Week_", week_counter,
"_Temperature(F)"))
  } else if (!is.na(types_row[i]) && grepl("Weather", types_row[i],
ignore.case = TRUE)) {
    col_names <- c(col_names, paste0("Week_", week_counter, "_Weather"))
    # Store the date for this week
    week_dates[paste0("Week_", week_counter)] <- dates_row[i - 1]
    week_counter <- week_counter + 1
  } else {
    col_names <- c(col_names, paste0("Column_", i))
  }
}

colnames(data) <- col_names

# Remove all-NA columns
data <- data %>% select(where(~!all(is.na(.))))

# Pivot longer
long_data <- data %>%
  pivot_longer(
    cols = -Location,
    names_to = c("Week", ".value"),
    names_pattern = "(Week_\\d+)_(.*)"
  ) %>%
  filter(!is.na(Location) & Location != "N/A")

# Add corresponding date for each week
```

```r
  long_data <- long_data %>%
    mutate(Date = week_dates[Week])

  # Convert Date to proper format and extract Year
  long_data <- long_data %>%
    mutate(
      Date = as.Date(Date, format = "%d-%b-%y"),
      Year = year(Date),
      Date = format(Date, "%d-%m-%Y")
    ) %>%
    select(Location, Year, Date, Week, "Temperature(F)", Weather)

  return(long_data)
}

# Apply function to all three datasets
weather_2022_long <- process_weather(weather_2022)
weather_2023_long <- process_weather(weather_2023)
weather_2024_long <- process_weather(weather_2024)%>%
  filter(Week != "Week_9")

# Combine into a single dataframe
all_weather_long <- bind_rows(weather_2022_long, weather_2023_long,
weather_2024_long) %>%
  rename(`Temperature (F)` = `Temperature(F)`) %>%
  mutate(
    `Temperature (F)` = as.numeric(`Temperature (F)`)
  )

# Define a named vector for mapping old names to new names
location_mapping <- c(
  "Living Stone" = "HOLS",
  "13th Street " = "13th Street",
  "Carondalet" = "Carondelet",
  "Carondelet " = "Carondelet",
  "HLS" = "HOLS",
  "Holy Cross " = "Holy Cross",
  "Orchard on Virginia" = "Virginia",
  "SLU Orchard" = "SLU",
  "Earthdance" = "EarthDance",
  "Earth Dance" = "EarthDance",
  "Florisant" = "Florissant",
  "Forissant" = "Florissant",
  "Kellog" = "Kellogg",
  "Kelloggg" = "Kellogg",
  "Kellogg " = "Kellogg",
  "Rustic Roots " = "Rustic Roots",
  "Emmanuel " = "Emmanuel",
  "Mckinley" = "McKinley"
```

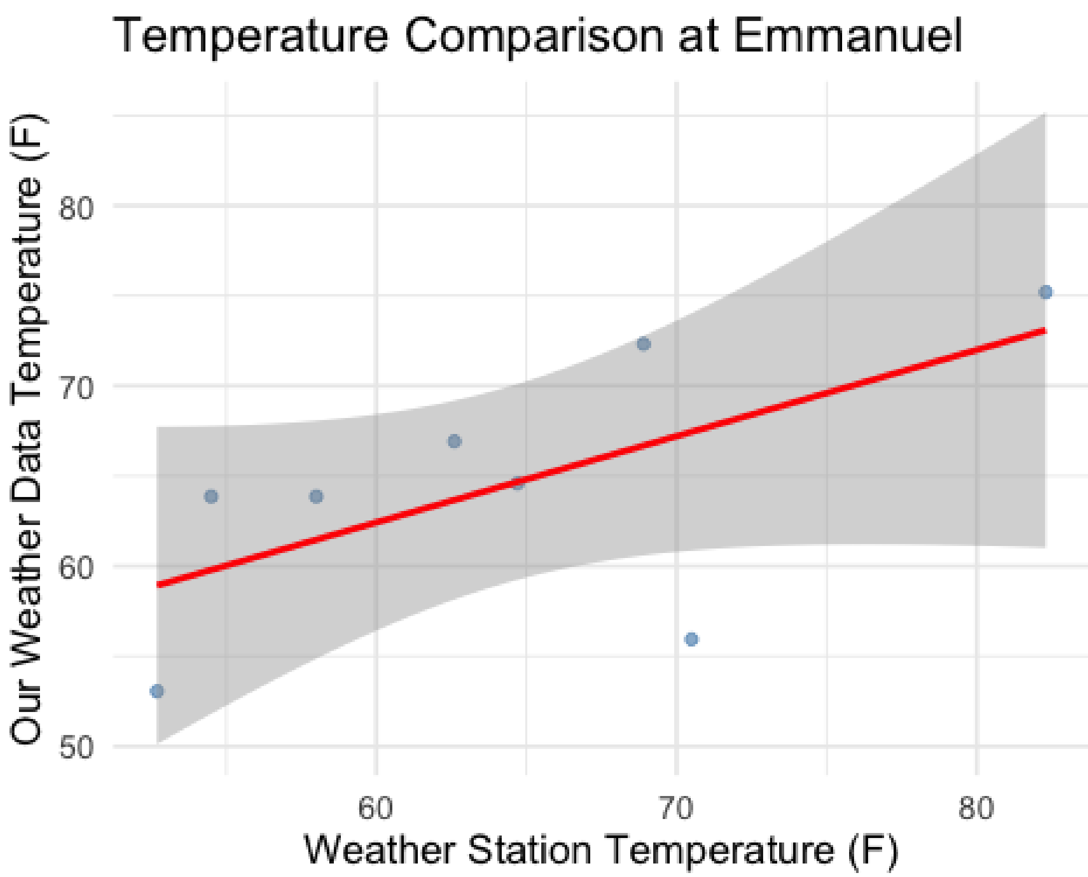
```r
)

# Rename locations in weather_file
weather_file <- weather_file %>%
  mutate(Location = recode(Location, !!!location_mapping))

# Rename locations in all_weather_long
all_weather_long <- all_weather_long %>%
  mutate(Location = recode(Location, !!!location_mapping))

# Removing NAs/blank spaces from Time
weather_file <- weather_file %>%
  filter(!is.na(Time) & Time != "")

# Merge dataset by Location and Date
#merged_weather <- bind_rows(all_weather_long, weather_file)
merged_weather <- full_join(weather_file, all_weatherdb,
  by = c("Date", "Hour (CST)", "Year"),
  suffix = c("", "_db")
)

# Removing NAs/blank spaces from Time
merged_weather <- merged_weather %>%
  filter(!is.na(Time) & Time != "") %>%
  select(Location, Year, Date, `Hour (CST)`, Time, everything())

# Ensure Location is a factor
merged_weather$Location <- as.factor(merged_weather$Location)

# Run linear regression for each Location
locations <- unique(merged_weather$Location)

for (loc in locations) {
  cat("\n====================\n")
  cat("Location:", loc, "\n")
  cat("====================\n")

  # Filter dataset for this location
  site_data <- merged_weather %>%
    filter(Location == loc)

  # --- Temperature comparison ---
  cat("\n--- Temperature Regression ---\n")
  temp_data <- site_data %>% select(`Temperature (F)`, `Temperature (F)_db`)
%>% na.omit()
  model_temp <- lm(`Temperature (F)` ~ `Temperature (F)_db`, data =
temp_data)
  print(summary(model_temp))
```

```r
  p_temp <- ggplot(temp_data, aes(x = `Temperature (F)_db`, y = `Temperature
(F)`)) +
    geom_point(alpha = 0.6, color = "steelblue") +
    geom_smooth(method = "lm", se = TRUE, color = "red") +
    labs(
      title = paste("Temperature Comparison at", loc),
      x = "Weather Station Temperature (F)",
      y = "Our Weather Data Temperature (F)"
    ) +
    theme_minimal(base_size = 13)
  print(p_temp)

  # --- Humidity comparison ---
  cat("\n--- Humidity Regression ---\n")
  hum_data <- site_data %>% select(`Humidity (%)`, `Humidity (%)_db`) %>%
na.omit()
  model_hum <- lm(`Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
  print(summary(model_hum))

  p_hum <- ggplot(hum_data, aes(x = `Humidity (%)_db`, y = `Humidity (%)`)) +
    geom_point(alpha = 0.6, color = "darkgreen") +
    geom_smooth(method = "lm", se = TRUE, color = "orange") +
    labs(
      title = paste("Humidity Comparison at", loc),
      x = "Weather Station Humidity (%)",
      y = "Our Weather Data Humidity (%)"
    ) +
    theme_minimal(base_size = 13)
  print(p_hum)

  # --- Wind Speed comparison ---
  cat("\n--- Wind Speed Regression ---\n")
  wind_data <- site_data %>% select(`WindSpeed (mph)`, `WindSpeed (mph)_db`)
%>% na.omit()
  model_wind <- lm(`WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data =
wind_data)
  print(summary(model_wind))

  p_wind <- ggplot(wind_data, aes(x = `WindSpeed (mph)_db`, y = `WindSpeed
(mph)`)) +
    geom_point(alpha = 0.6, color = "purple") +
    geom_smooth(method = "lm", se = TRUE, color = "red") +
    labs(
      title = paste("Wind Speed Comparison at", loc),
      x = "Weather Station Wind Speed (mph)",
      y = "Our Weather Data Wind Speed (mph)"
    ) +
    theme_minimal(base_size = 13)
```

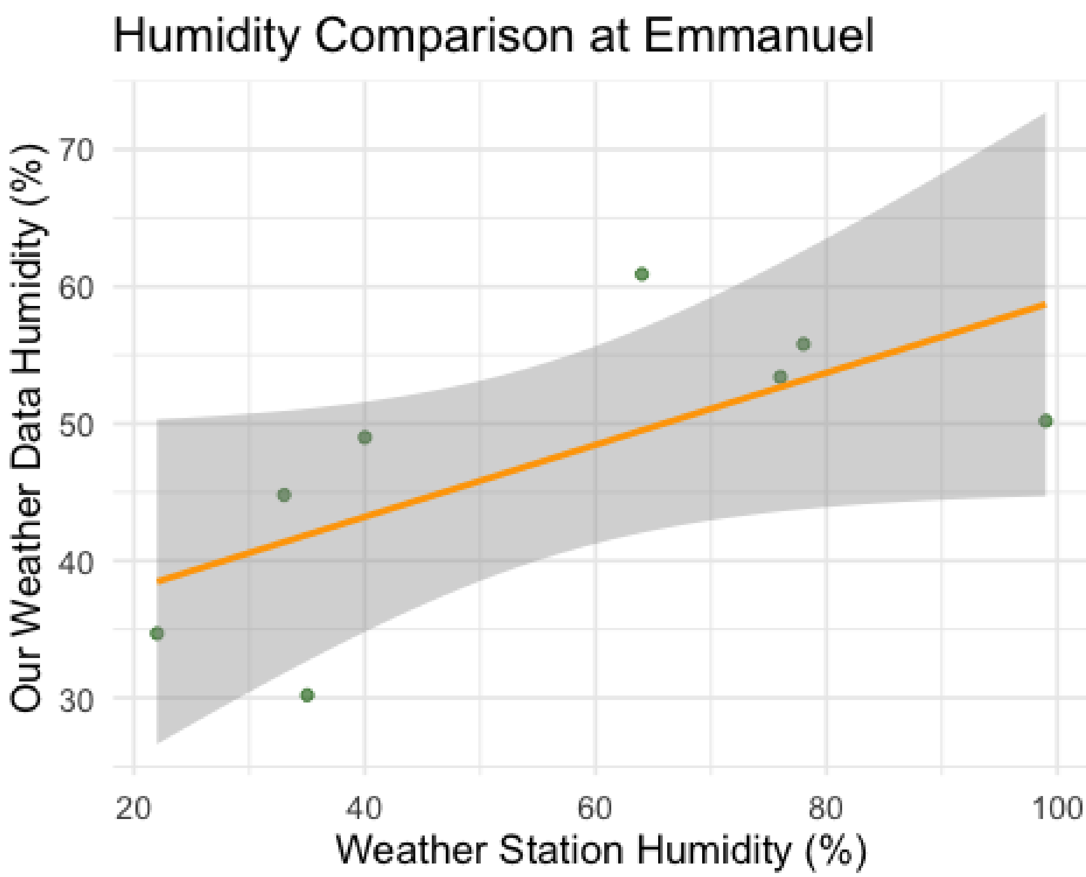
```
  print(p_wind)
}
```

```
##
## ====================
## Location: Emmanuel
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.507  -1.535   2.251   3.458   5.639
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           33.7045    15.8823   2.122   0.0781 .
## `Temperature (F)_db`   0.4786     0.2447   1.956   0.0982 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.271 on 6 degrees of freedom
## Multiple R-squared:  0.3894, Adjusted R-squared:  0.2876
## F-statistic: 3.826 on 1 and 6 DF,  p-value: 0.09823
```

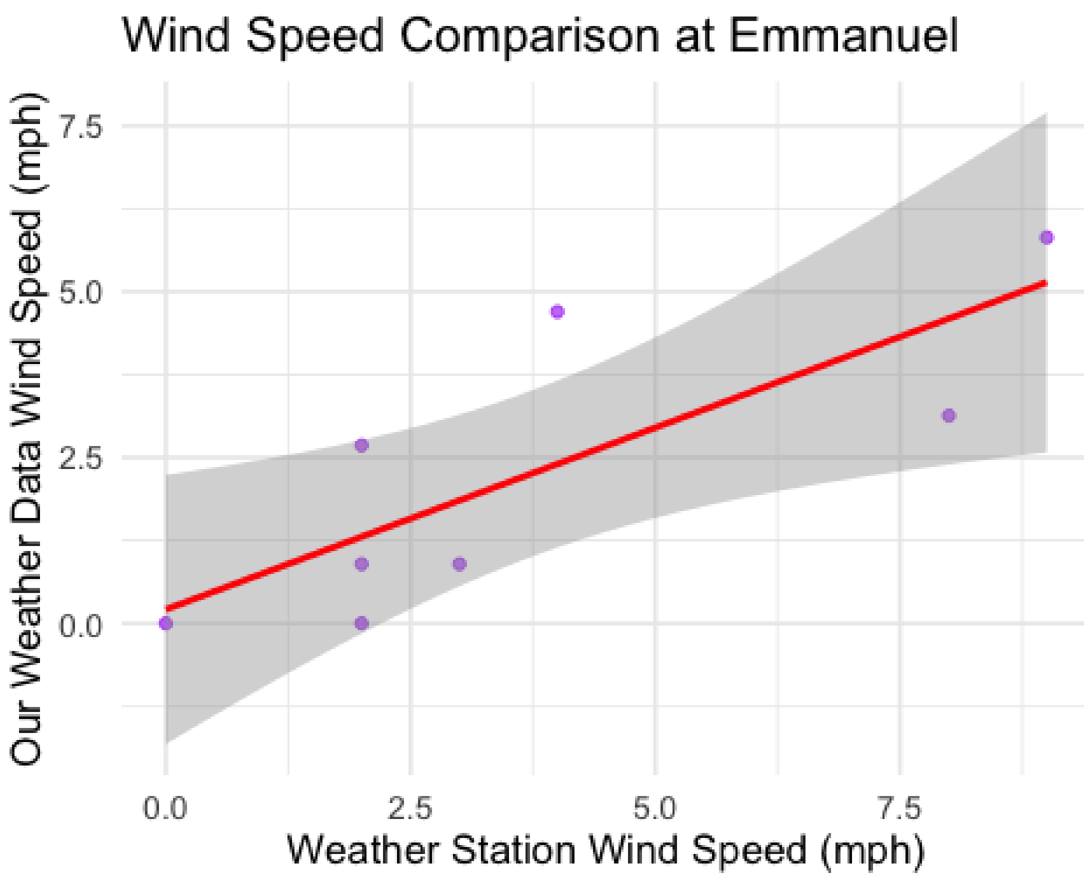
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.692  -4.958   1.676   4.024  11.391
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         32.6979     7.0157   4.661  0.00347 **
## `Humidity (%)_db`    0.2627     0.1143   2.299  0.06121 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.223 on 6 degrees of freedom
## Multiple R-squared:  0.4683, Adjusted R-squared:  0.3797
## F-statistic: 5.284 on 1 and 6 DF,  p-value: 0.06121

## `geom_smooth()` using formula = 'y ~ x'
```

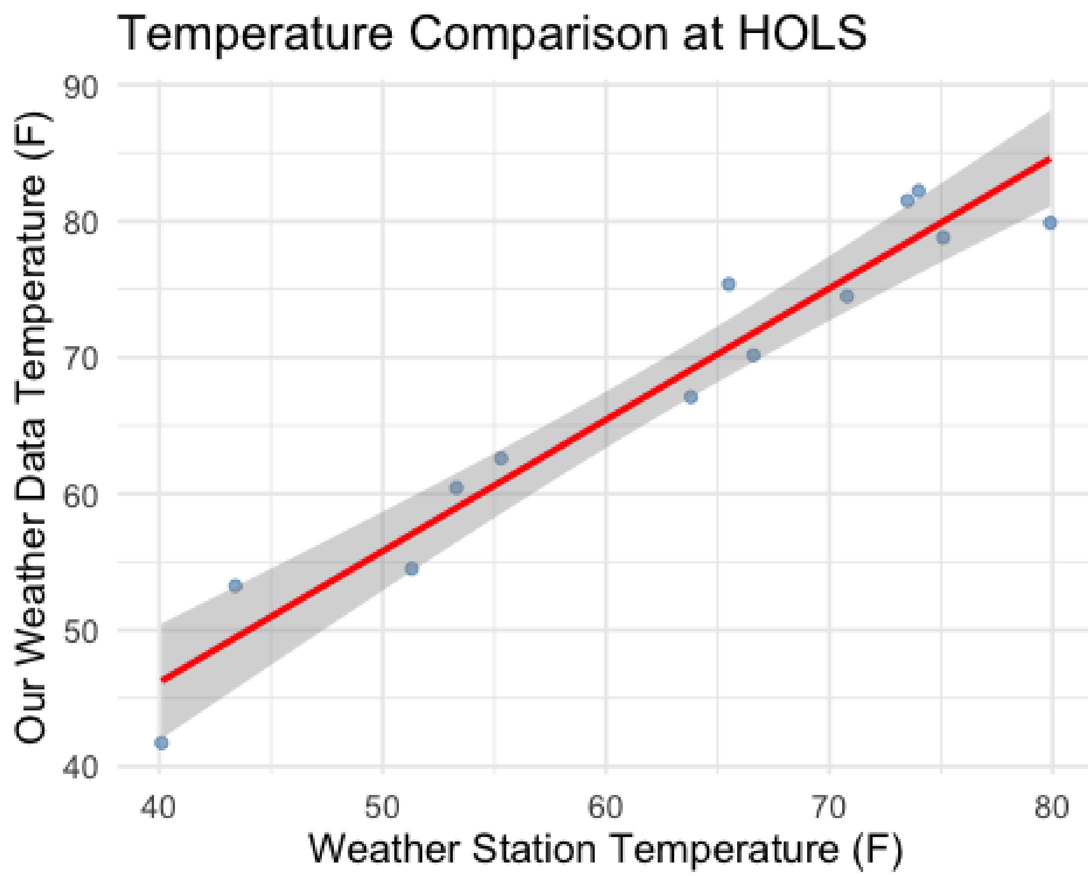



```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.4621 -1.0458 -0.3106  0.8503  2.2957
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            0.2100     0.8291   0.253   0.8085
## `WindSpeed (mph)_db`   0.5480     0.1738   3.152   0.0198 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.449 on 6 degrees of freedom
## Multiple R-squared:  0.6235, Adjusted R-squared:  0.5608
## F-statistic: 9.937 on 1 and 6 DF,  p-value: 0.01976

## `geom_smooth()` using formula = 'y ~ x'
```



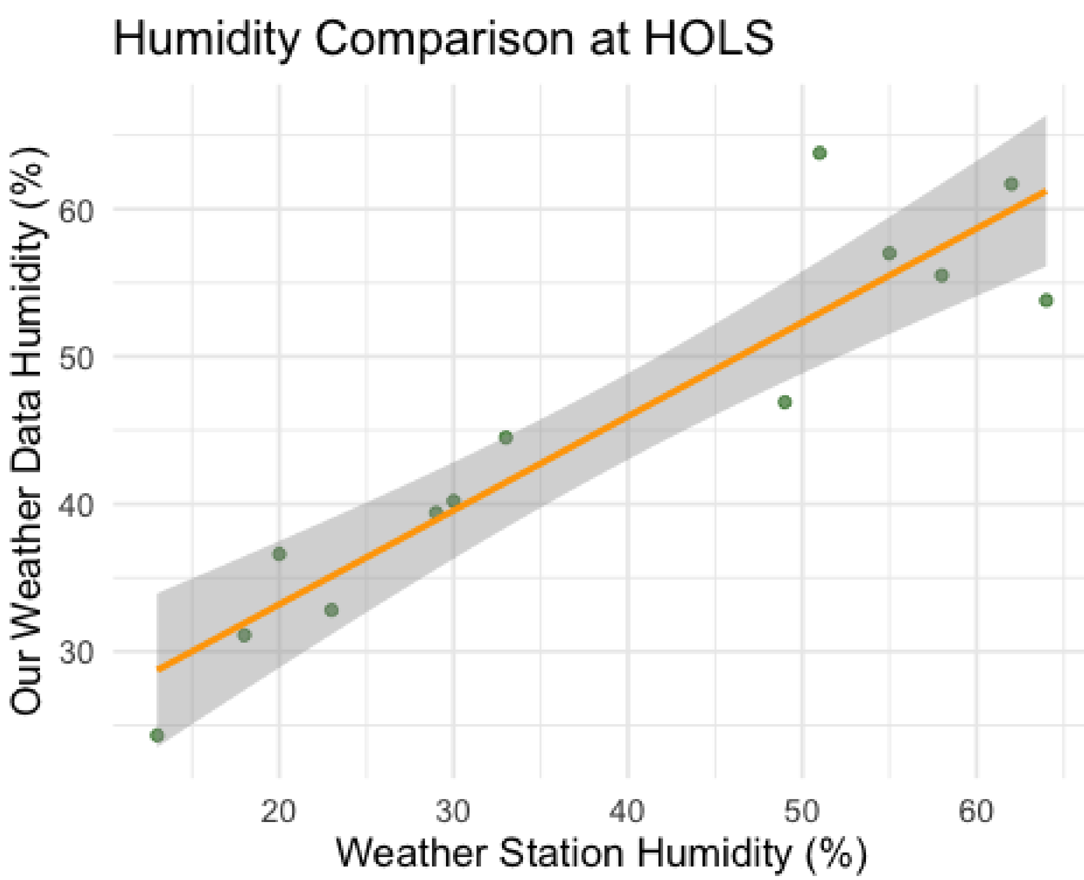

```
##
## ====================
## Location: HOLS
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -4.723 -1.993 -1.179  3.062  4.649
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           7.62749    4.78912   1.593     0.14
## `Temperature (F)_db`  0.96340    0.07519  12.812 5.92e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.312 on 11 degrees of freedom
```

```
## Multiple R-squared:  0.9372, Adjusted R-squared:  0.9315
## F-statistic: 164.2 on 1 and 11 DF,  p-value: 5.917e-08

## `geom_smooth()` using formula = 'y ~ x'
```



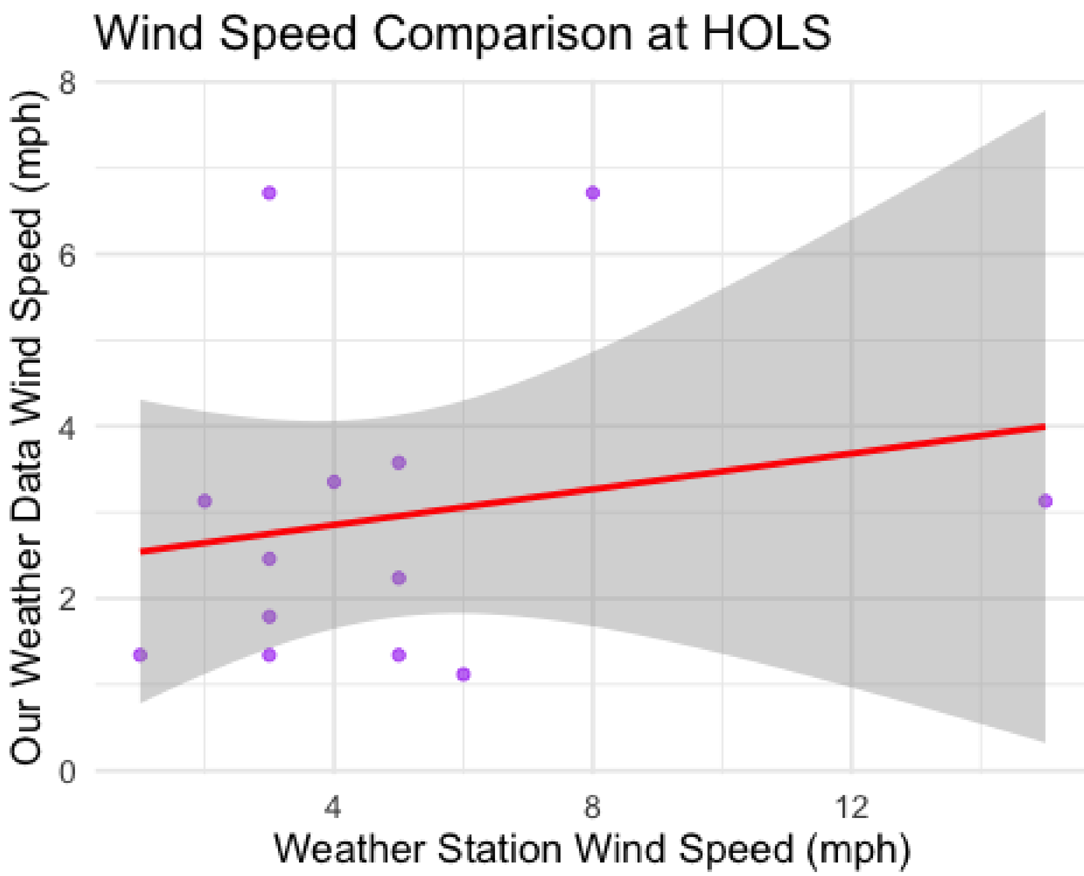

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.4329 -2.2998  0.4759  1.7419 10.8532
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       20.43978    3.23545   6.317 5.70e-05 ***
## `Humidity (%)_db`  0.63739    0.07596   8.391 4.14e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.784 on 11 degrees of freedom
```

```
## Multiple R-squared:  0.8649, Adjusted R-squared:  0.8526
## F-statistic:  70.4 on 1 and 11 DF,  p-value: 4.139e-06

## `geom_smooth()` using formula = 'y ~ x'
```



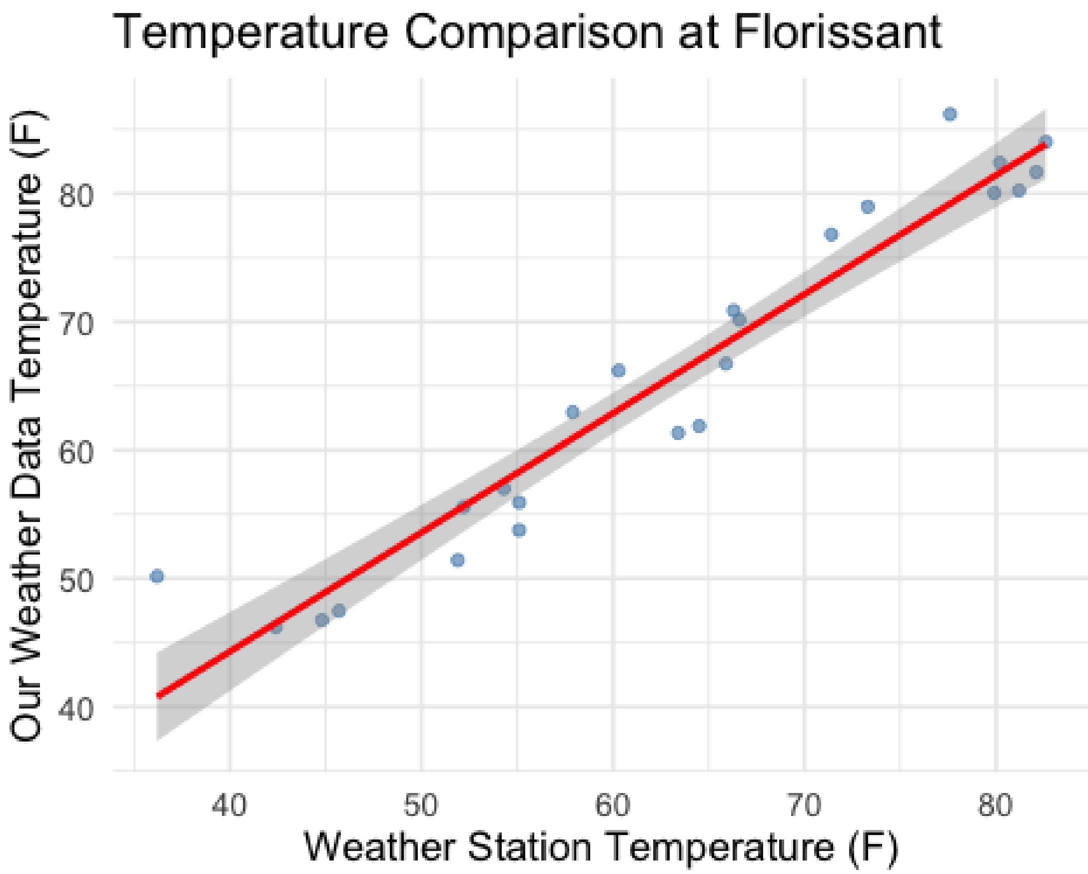

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9436 -1.2015 -0.7214  0.5007  3.9598
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            2.4400     0.9236   2.642   0.0229 *
## `WindSpeed (mph)_db`   0.1037     0.1558   0.666   0.5194
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.918 on 11 degrees of freedom
```

```
## Multiple R-squared:  0.03872,    Adjusted R-squared:  -0.04867
## F-statistic: 0.443 on 1 and 11 DF,  p-value: 0.5194

## `geom_smooth()` using formula = 'y ~ x'
```



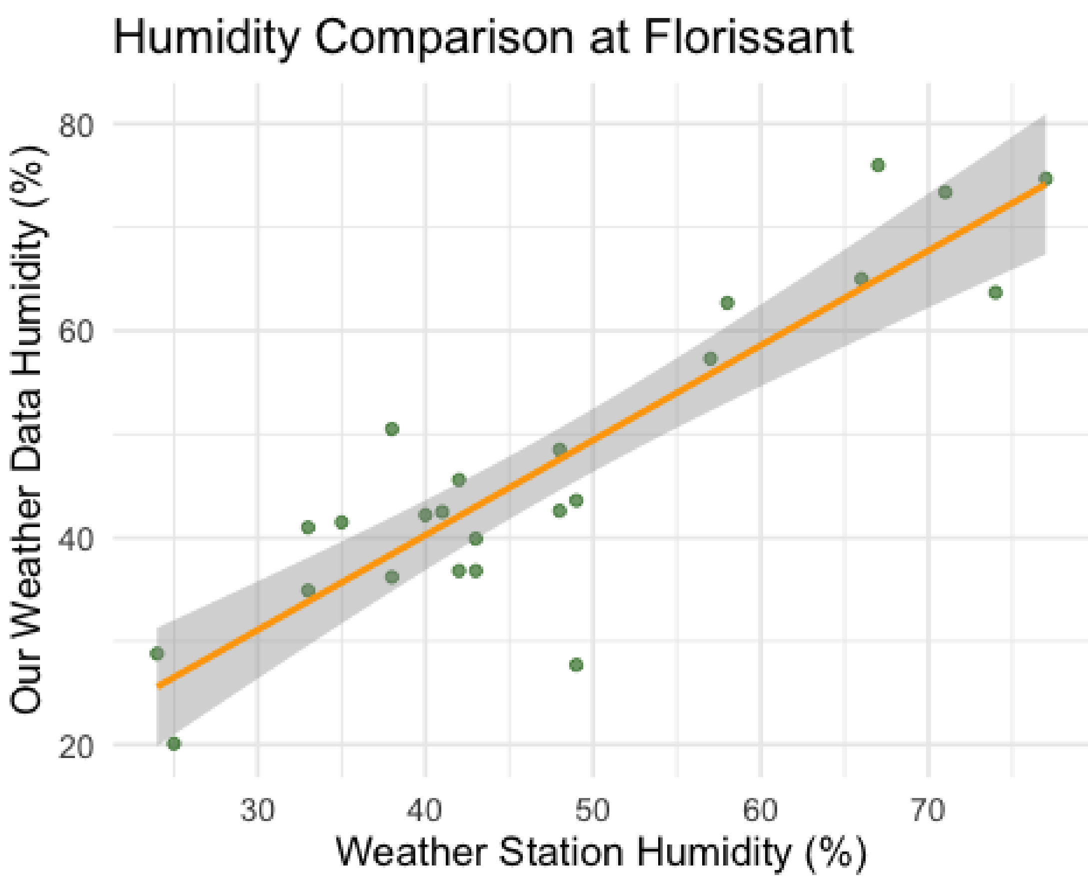

```
##
## ====================
## Location: Florissant
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.1770 -2.1769 -0.4503  2.0589  9.3846
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           7.20294    3.57665   2.014   0.0564 .
## `Temperature (F)_db`  0.92797    0.05556  16.701 5.55e-14 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.657 on 22 degrees of freedom
## Multiple R-squared:  0.9269, Adjusted R-squared:  0.9236
## F-statistic: 278.9 on 1 and 22 DF,  p-value: 5.549e-14

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.8036  -4.9244   0.9888   3.8172  12.0806
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         3.58301    4.95268   0.723    0.477
## `Humidity (%)_db`   0.91675    0.09963   9.202 5.36e-09 ***
```

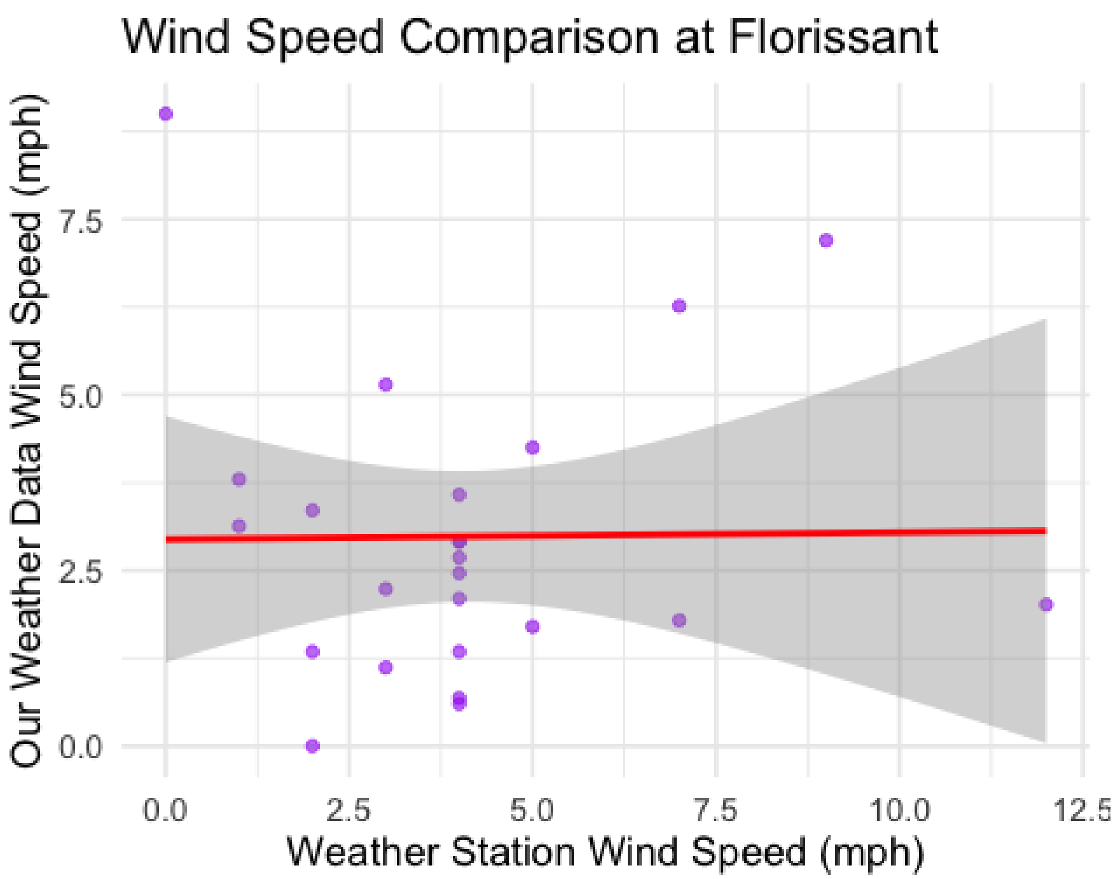
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.092 on 22 degrees of freedom
## Multiple R-squared:  0.7938, Adjusted R-squared:  0.7844
## F-statistic: 84.68 on 1 and 22 DF,  p-value: 5.362e-09

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.9633 -1.3749 -0.4105  0.6587  6.0565
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          2.943554   0.845021   3.483  0.00211 **
## `WindSpeed (mph)_db` 0.009863   0.175249   0.056  0.95563
```

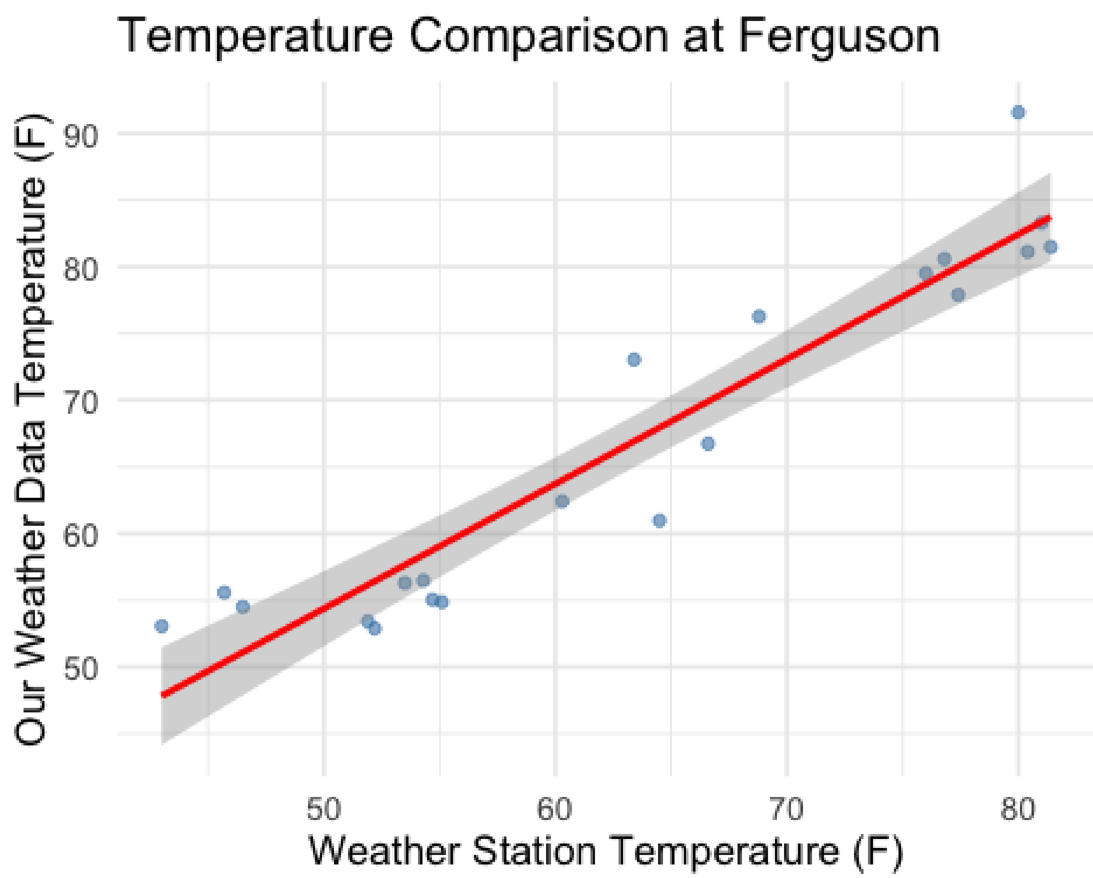
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.202 on 22 degrees of freedom
## Multiple R-squared:  0.000144,	Adjusted R-squared:  -0.0453
## F-statistic: 0.003168 on 1 and 22 DF,  p-value: 0.9556
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



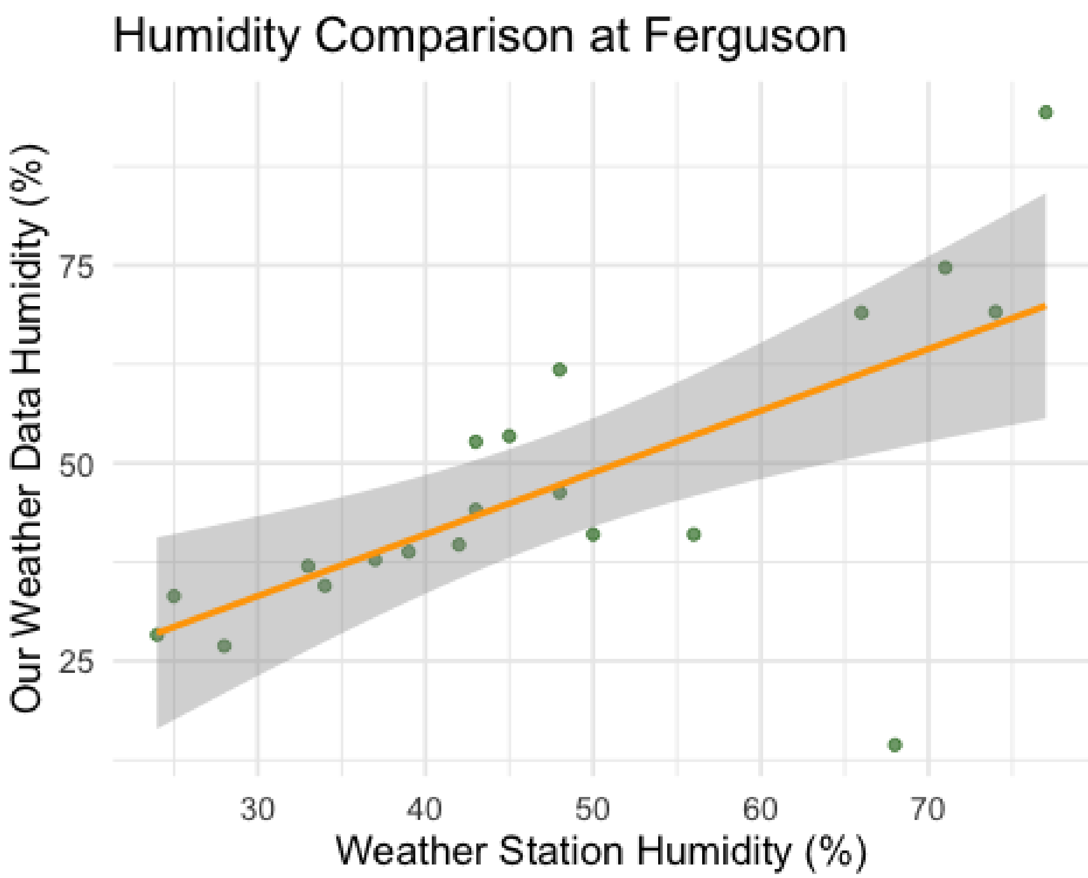

```
##
## ====================
## Location: Ferguson
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.962 -2.729 -1.591  3.405  9.131
##
```

```
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            7.57262    4.70566   1.609    0.124
## `Temperature (F)_db`   0.93595    0.07267  12.880 7.77e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.225 on 19 degrees of freedom
## Multiple R-squared:  0.8972, Adjusted R-squared:  0.8918
## F-statistic: 165.9 on 1 and 19 DF,  p-value: 7.773e-11

## `geom_smooth()` using formula = 'y ~ x'
```



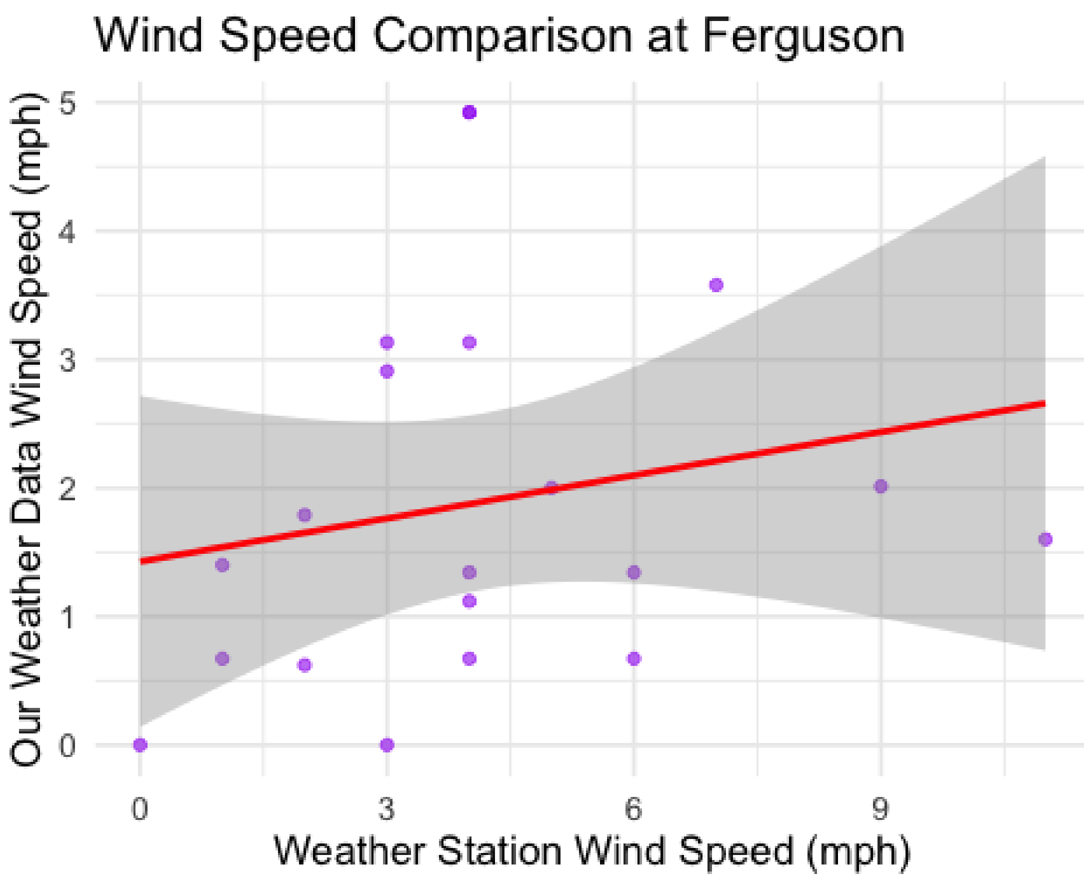

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -48.455  -2.089   0.261   7.902  24.424
##
```

```
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)          9.8027    10.1637   0.964  0.34758
## `Humidity (%)_db`    0.7802     0.2027   3.849  0.00118 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.42 on 18 degrees of freedom
## Multiple R-squared:  0.4514, Adjusted R-squared:  0.4209
## F-statistic: 14.81 on 1 and 18 DF,  p-value: 0.001177

## `geom_smooth()` using formula = 'y ~ x'
```



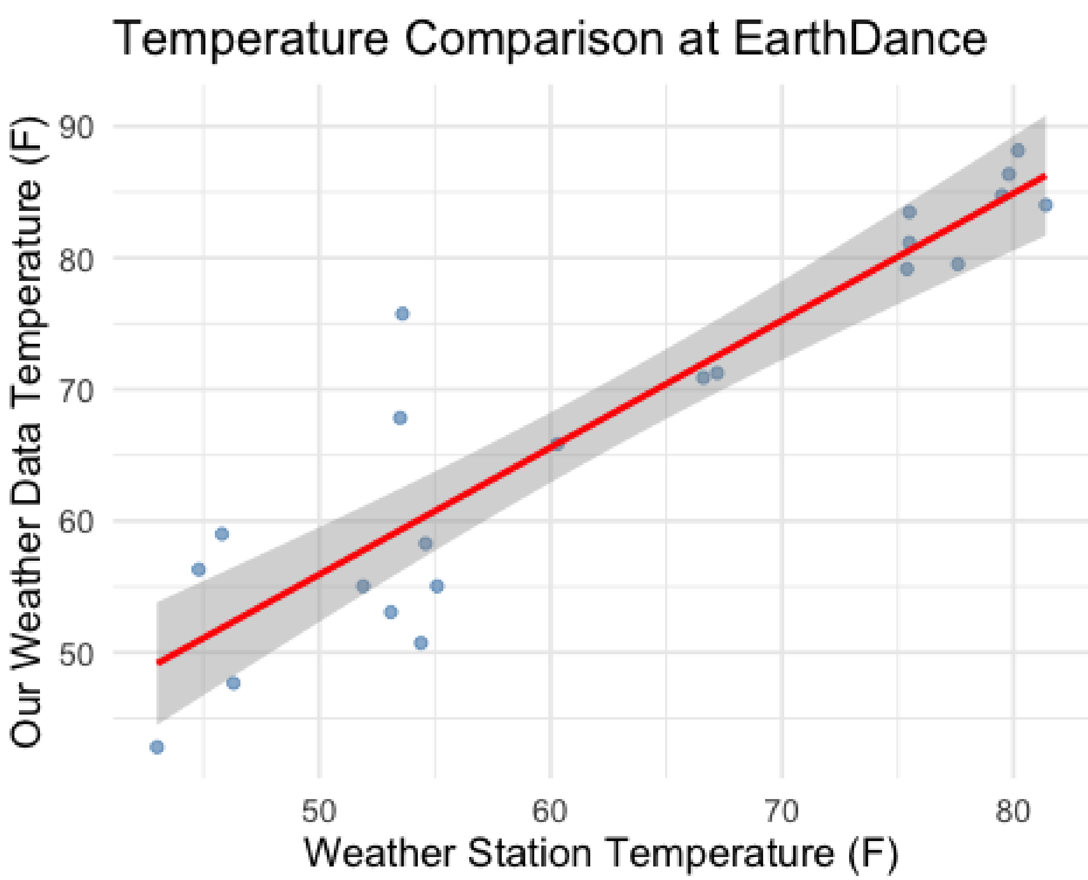

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7630 -1.0368 -0.4769  1.1730  3.0464
##
```

```
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.4273     0.6130   2.328   0.0318 *
## `WindSpeed (mph)_db`   0.1119     0.1250   0.895   0.3825
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.461 on 18 degrees of freedom
## Multiple R-squared:  0.04262,    Adjusted R-squared:  -0.01057
## F-statistic: 0.8013 on 1 and 18 DF,  p-value: 0.3825

## `geom_smooth()` using formula = 'y ~ x'
```



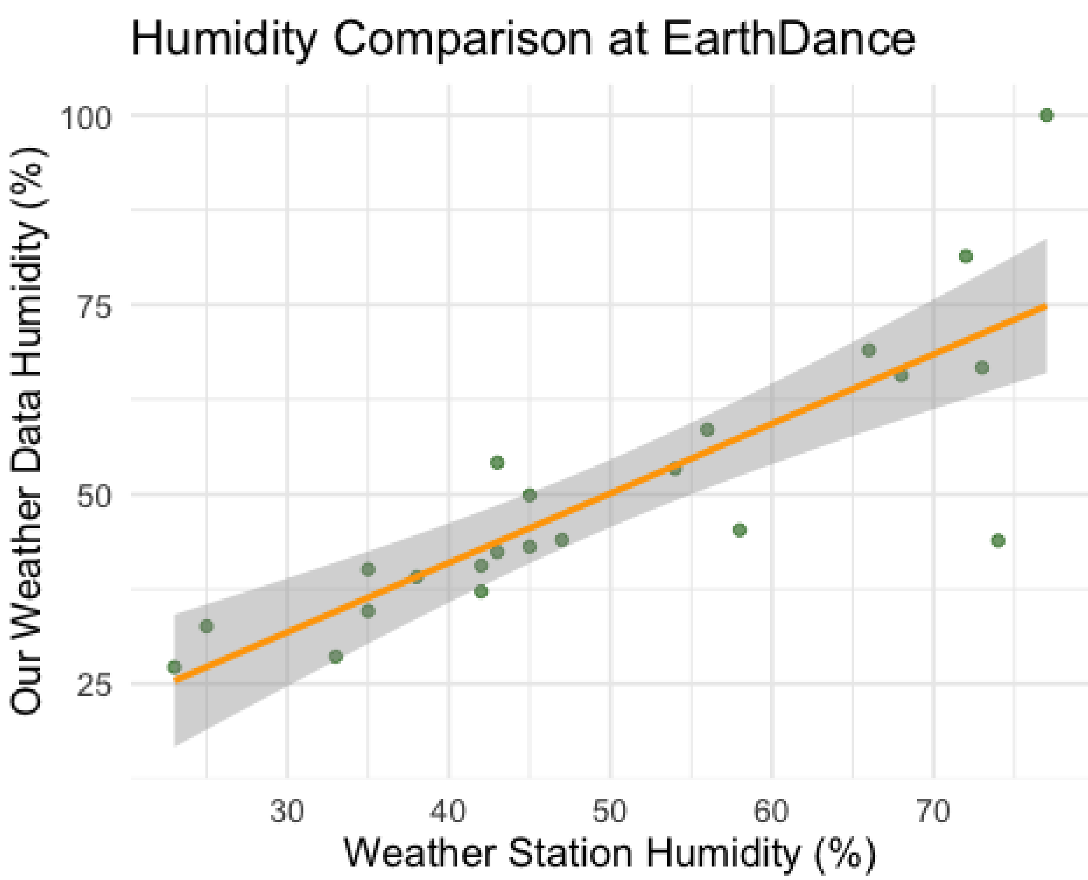

```
##
## ====================
## Location: EarthDance
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -9.448 -2.978 -1.187  2.603 16.345 
## 
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)    
## (Intercept)             7.6007     6.0808    1.25    0.226    
## `Temperature (F)_db`    0.9663     0.0952   10.15 2.46e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 5.879 on 20 degrees of freedom
## Multiple R-squared:  0.8374, Adjusted R-squared:  0.8293 
## F-statistic:   103 on 1 and 20 DF,  p-value: 2.459e-09

## `geom_smooth()` using formula = 'y ~ x'
```



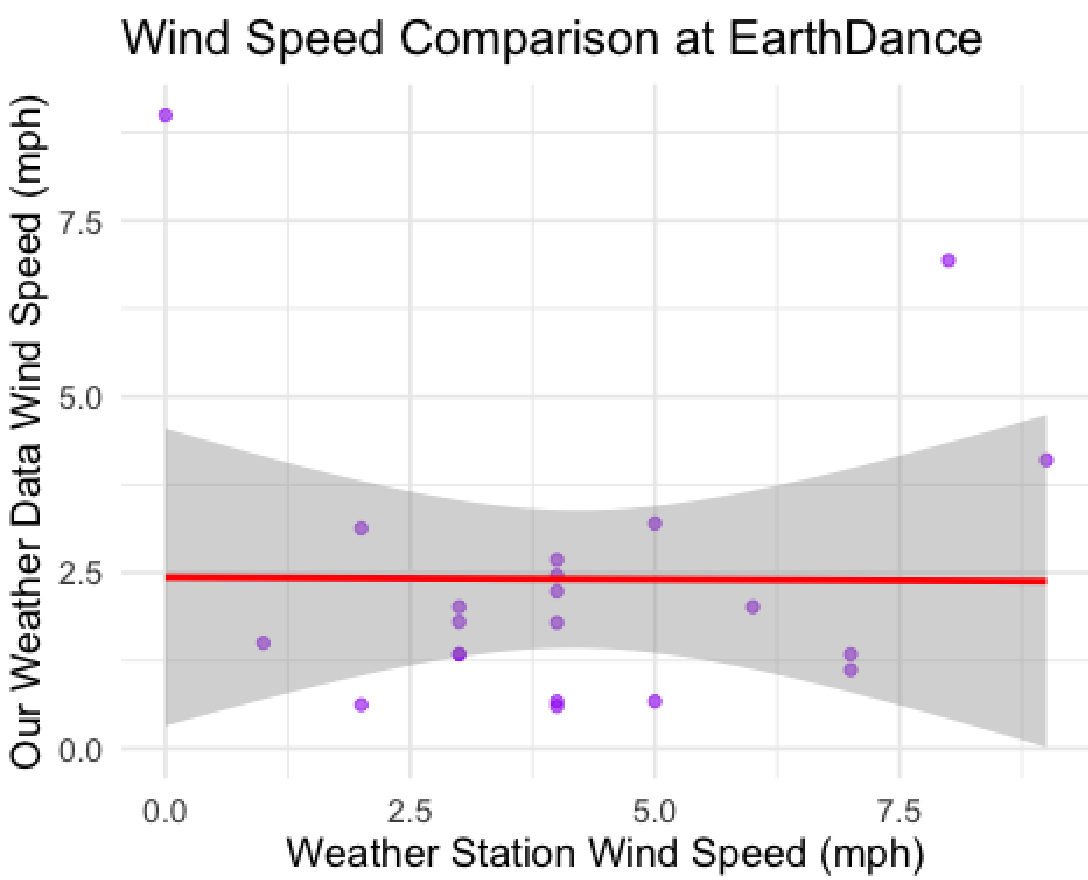

```
## 
## --- Humidity Regression ---
## 
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
## 
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max 
## -28.2247  -3.1546  -0.6643   4.0802  25.1267 
## 
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)    
## (Intercept)         4.3268     7.0431   0.614    0.546    
## `Humidity (%)_db`   0.9162     0.1349   6.790 1.33e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 10.04 on 20 degrees of freedom
## Multiple R-squared:  0.6974, Adjusted R-squared:  0.6823 
## F-statistic:  46.1 on 1 and 20 DF,  p-value: 1.331e-06

## `geom_smooth()` using formula = 'y ~ x'
```



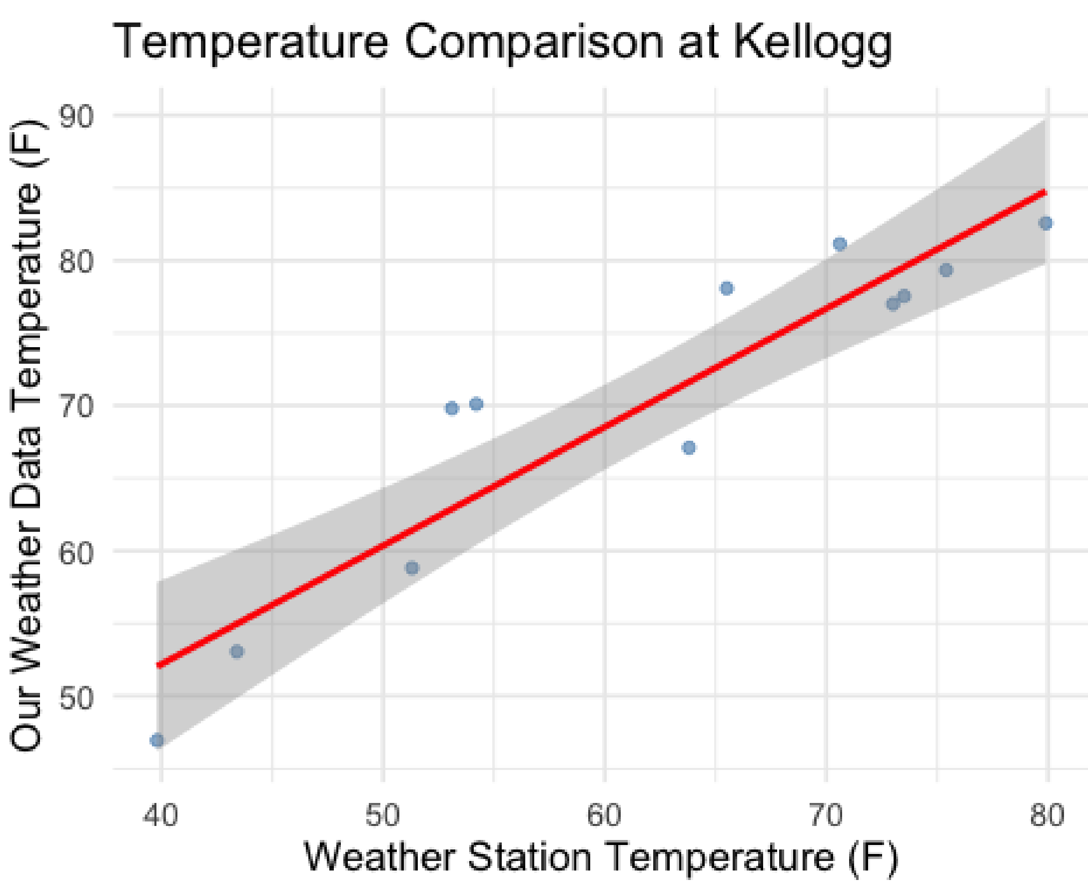

```
## 
## --- Wind Speed Regression ---
## 
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
## 
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8094 -1.0733 -0.6155  0.2749  6.5662
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)             2.433852   1.009102   2.412   0.0261 *
## `WindSpeed (mph)_db`   -0.006116   0.213302  -0.029   0.9774
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.146 on 19 degrees of freedom
## Multiple R-squared:  4.326e-05,  Adjusted R-squared:  -0.05259
## F-statistic: 0.0008221 on 1 and 19 DF,  p-value: 0.9774

## `geom_smooth()` using formula = 'y ~ x'
```



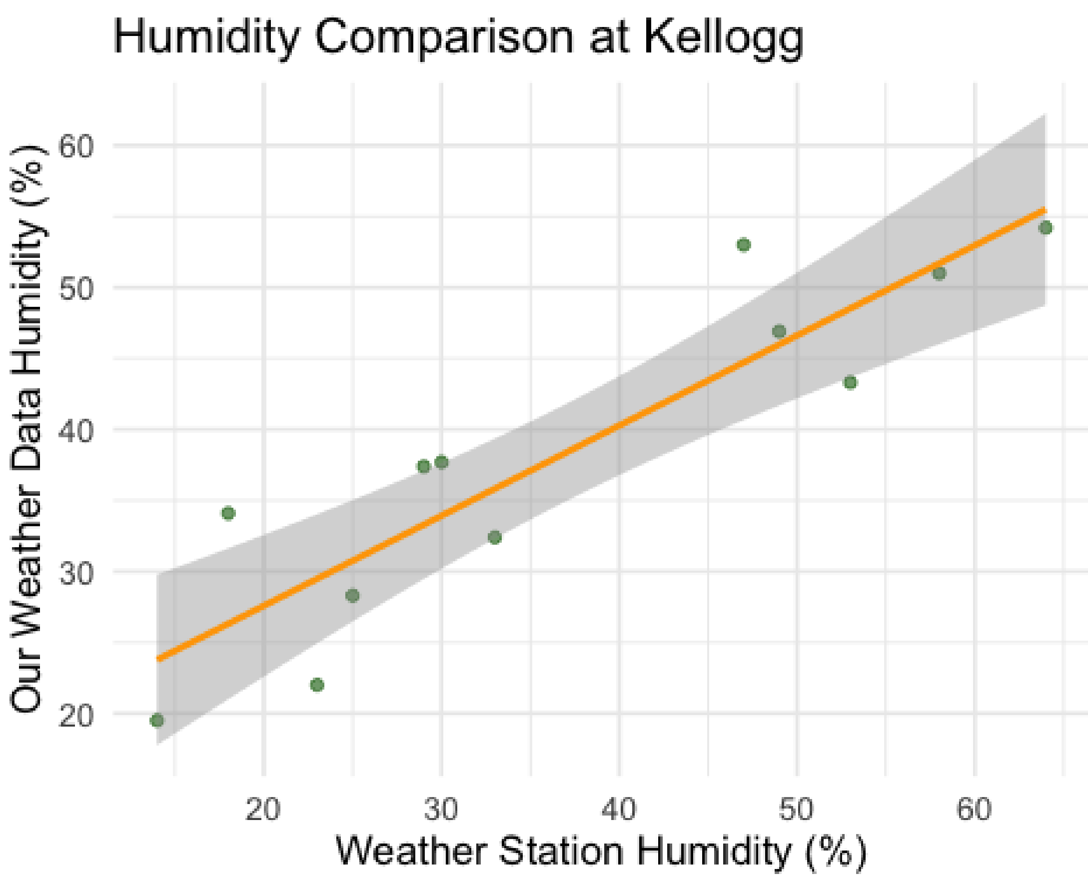

```
##
## ====================
## Location: Kellogg
## ====================
##
## --- Temperature Regression ---
```

```
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -5.095 -2.296 -1.961  4.235  6.907
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           19.5418     6.4656   3.022   0.0128 *
## `Temperature (F)_db`   0.8164     0.1022   7.986  1.2e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.498 on 10 degrees of freedom
## Multiple R-squared:  0.8645, Adjusted R-squared:  0.8509
## F-statistic: 63.78 on 1 and 10 DF,  p-value: 1.195e-05

## `geom_smooth()` using formula = 'y ~ x'
```



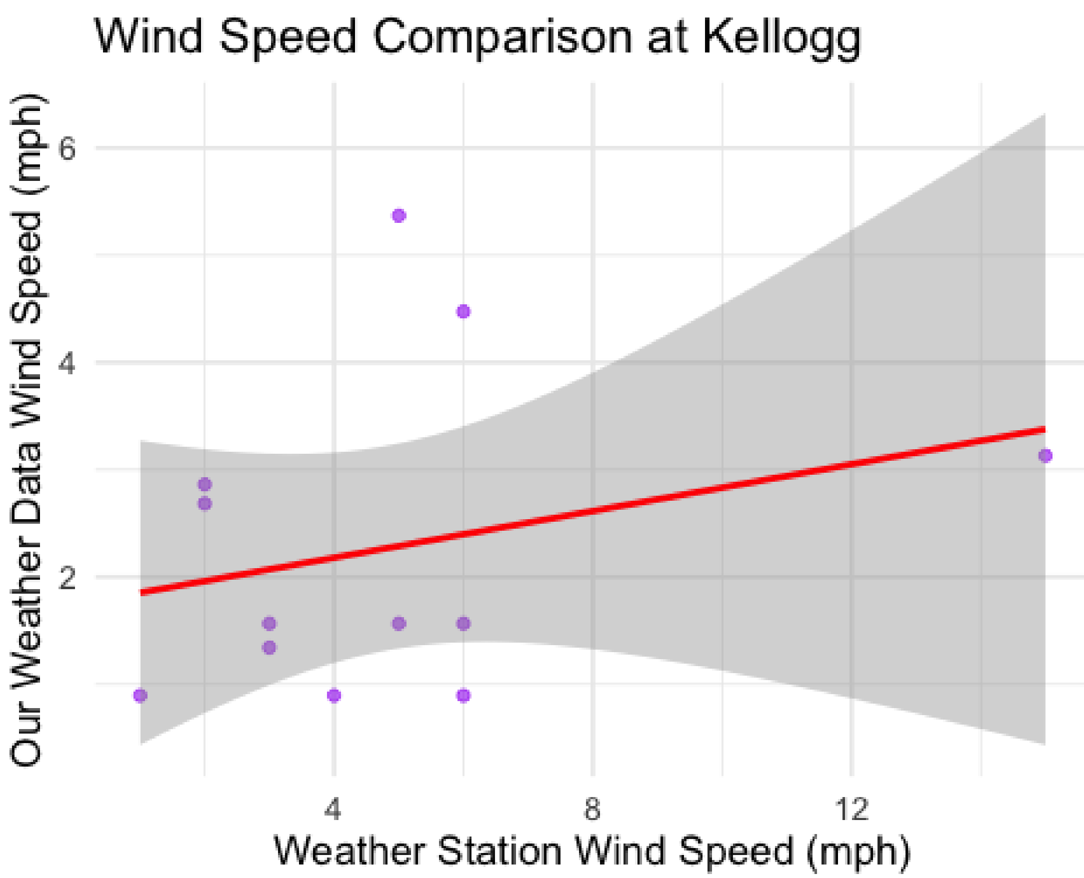

```
##
## --- Humidity Regression ---
```

```
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.4876 -3.6433 -0.9957  3.8550  8.2862
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       14.89586    3.88328   3.836  0.00329 **
## `Humidity (%)_db`  0.63442    0.09662   6.566 6.34e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.318 on 10 degrees of freedom
## Multiple R-squared:  0.8117, Adjusted R-squared:  0.7929
## F-statistic: 43.11 on 1 and 10 DF,  p-value: 6.344e-05

## `geom_smooth()` using formula = 'y ~ x'
```



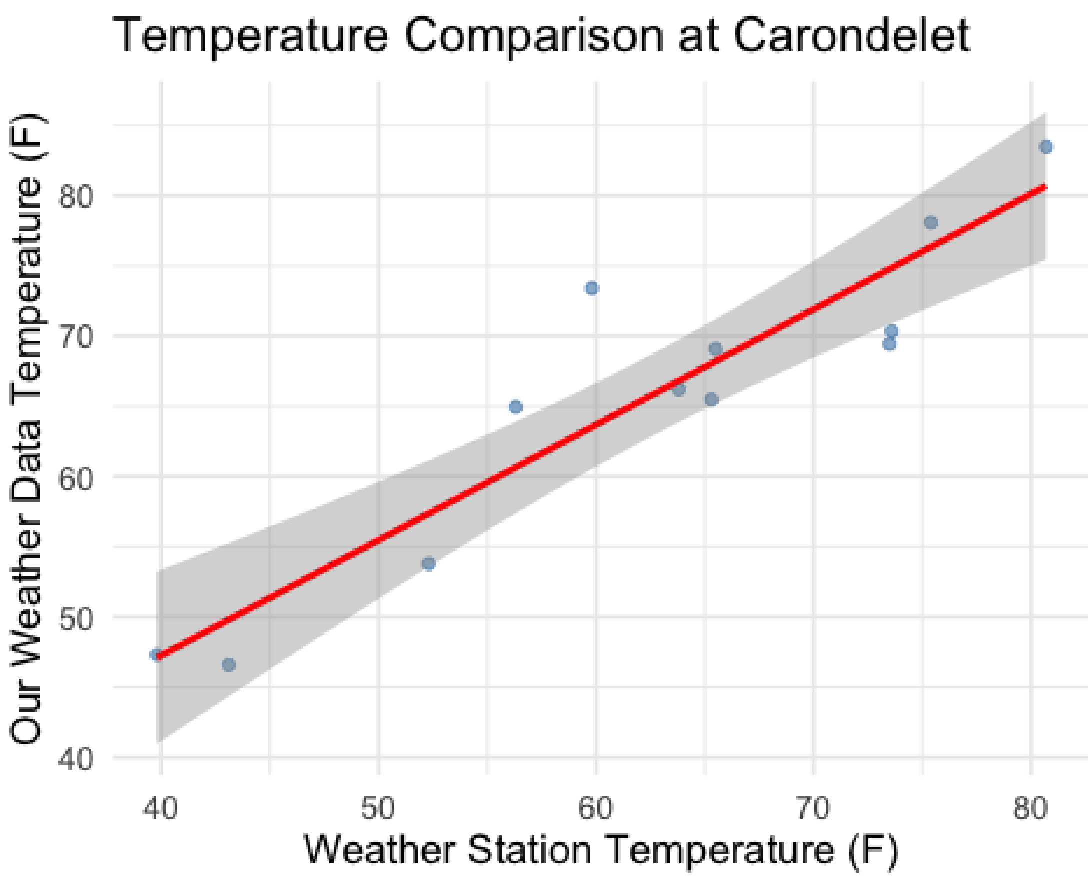

```
##
## --- Wind Speed Regression ---
```

```
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5027 -0.8633 -0.6139  0.7671  3.0800
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.7443     0.7326   2.381   0.0385 *
## `WindSpeed (mph)_db`   0.1089     0.1230   0.886   0.3967
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.484 on 10 degrees of freedom
## Multiple R-squared:  0.07272,    Adjusted R-squared:  -0.02001
## F-statistic: 0.7842 on 1 and 10 DF,  p-value: 0.3967

## `geom_smooth()` using formula = 'y ~ x'
```

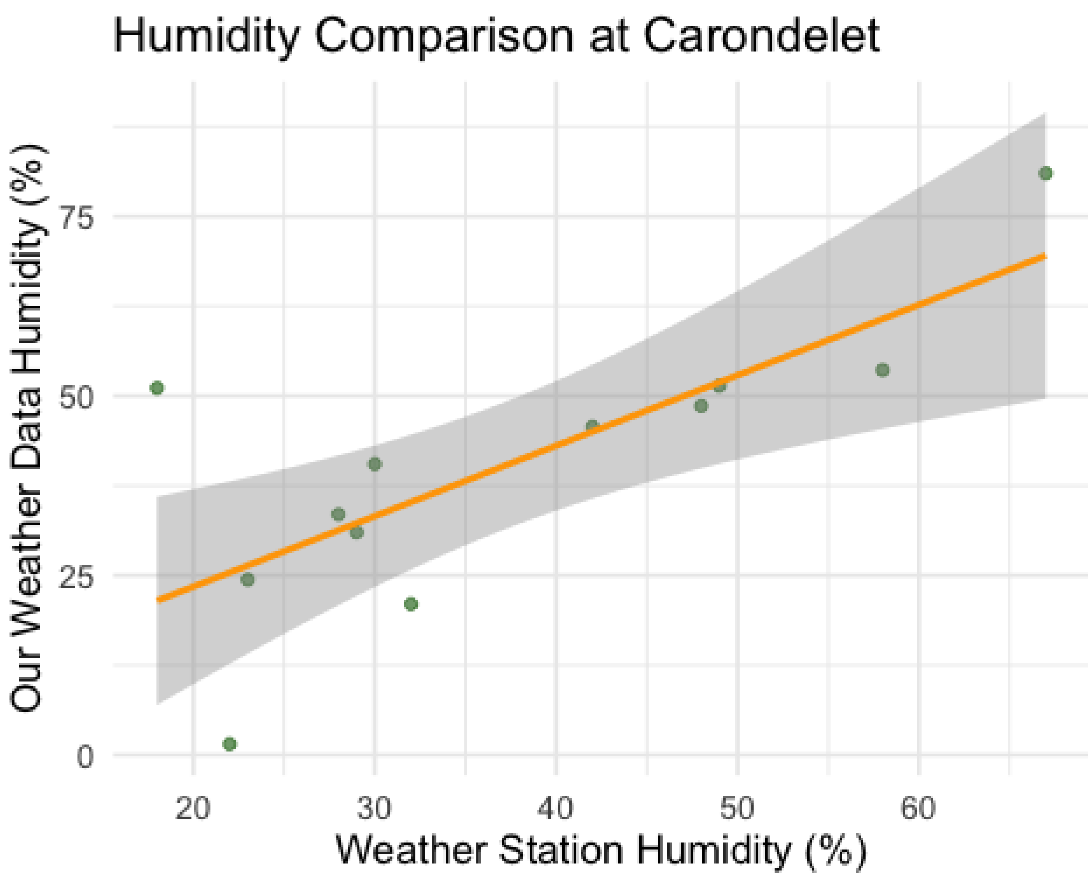



```
##
## ====================
```

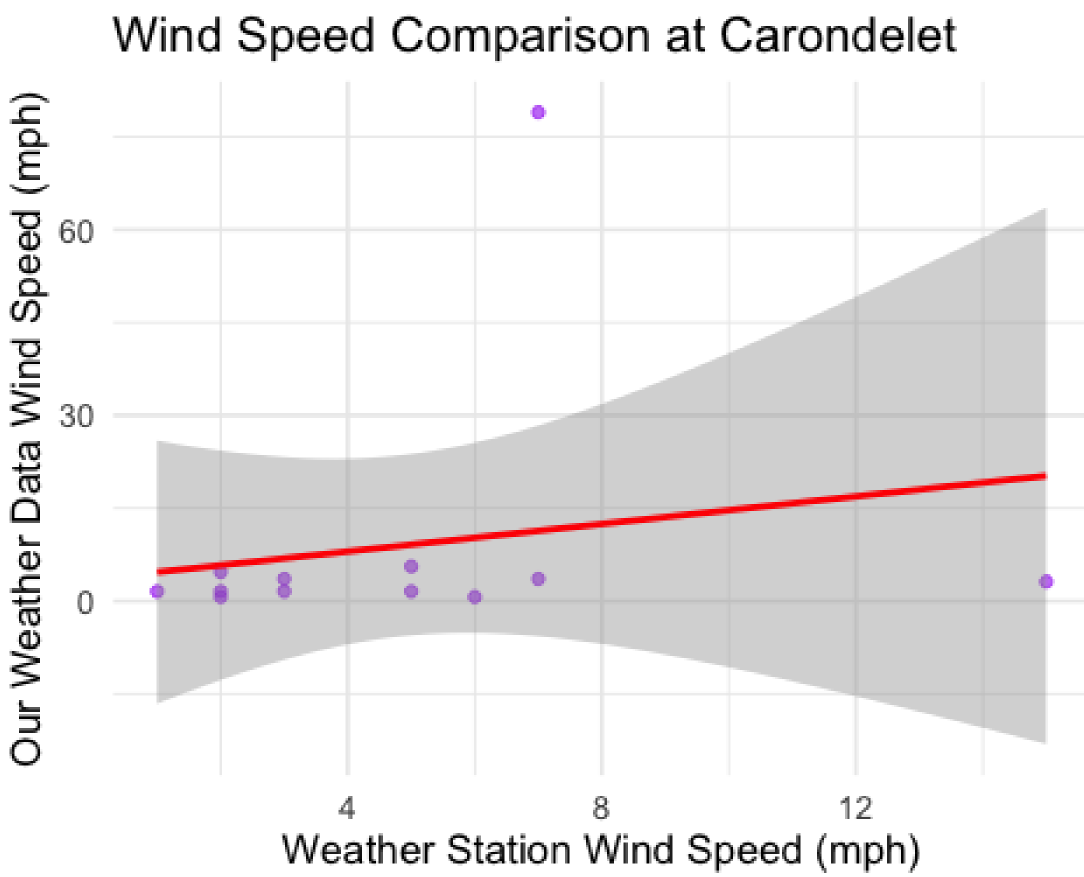
```
## Location: Carondelet
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.3336 -3.3082 -0.1961  2.0069  9.8815
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           14.3902     6.8132   2.112   0.0608 .
## `Temperature (F)_db`   0.8215     0.1071   7.671  1.7e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.546 on 10 degrees of freedom
## Multiple R-squared:  0.8547, Adjusted R-squared:  0.8402
## F-statistic: 58.84 on 1 and 10 DF,  p-value: 1.698e-05

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.8949  -3.5064  -0.8737   3.4779  29.6296
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)          3.8105    10.7546   0.354  0.73046
## `Humidity (%)_db`    0.9811     0.2688   3.650  0.00446 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.79 on 10 degrees of freedom
## Multiple R-squared:  0.5712, Adjusted R-squared:  0.5283
## F-statistic: 13.32 on 1 and 10 DF,  p-value: 0.004464

## `geom_smooth()` using formula = 'y ~ x'
```

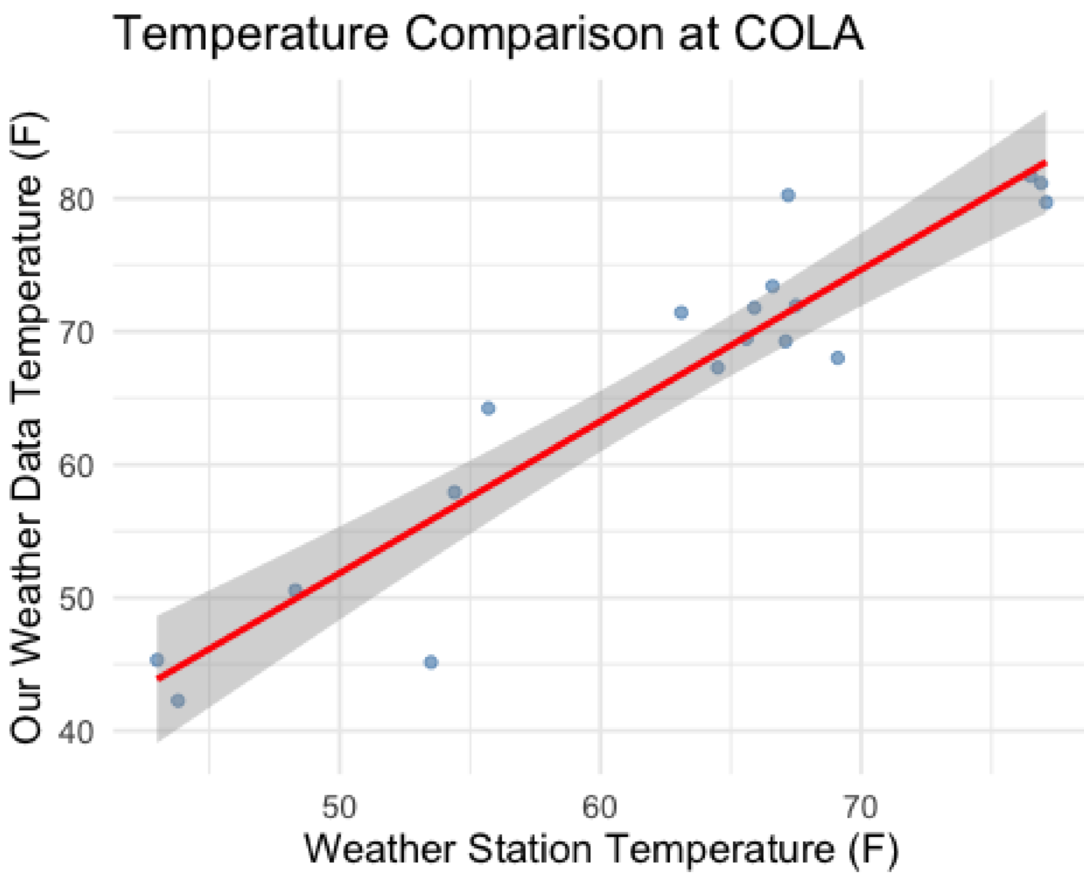



```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.104  -7.601  -4.660  -3.258  67.625
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)             3.554     10.907   0.326    0.751
## `WindSpeed (mph)_db`    1.112      1.801   0.617    0.551
##
## Residual standard error: 22.76 on 10 degrees of freedom
## Multiple R-squared:  0.03672,	Adjusted R-squared:  -0.05961
## F-statistic: 0.3812 on 1 and 10 DF,  p-value: 0.5507

## `geom_smooth()` using formula = 'y ~ x'
```



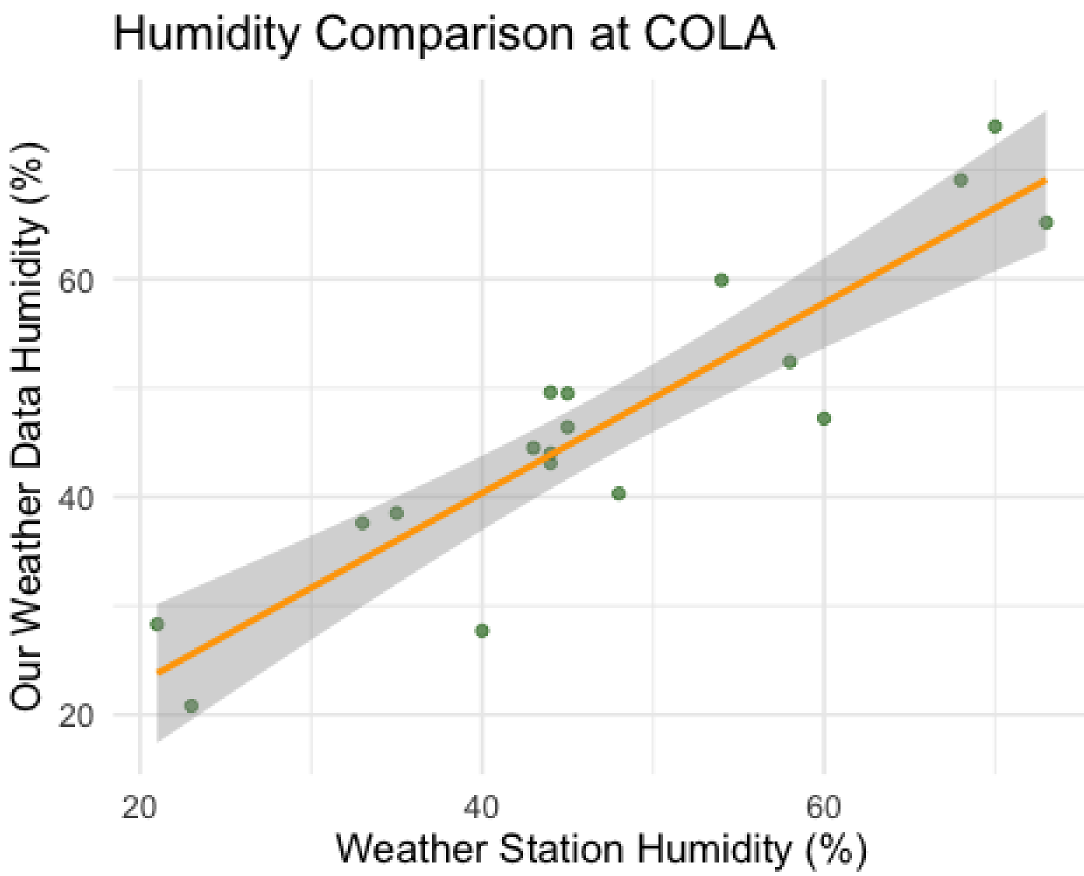

```
##
## ====================
## Location: COLA
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6971  -1.9081  -0.0173   1.7168   8.7815
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -5.1664     6.4729  -0.798    0.436
## `Temperature (F)_db`   1.1403     0.1021  11.165 5.81e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.446 on 16 degrees of freedom
```

```
## Multiple R-squared:  0.8862, Adjusted R-squared:  0.8791 
## F-statistic: 124.7 on 1 and 16 DF,  p-value: 5.808e-09

## `geom_smooth()` using formula = 'y ~ x'
```



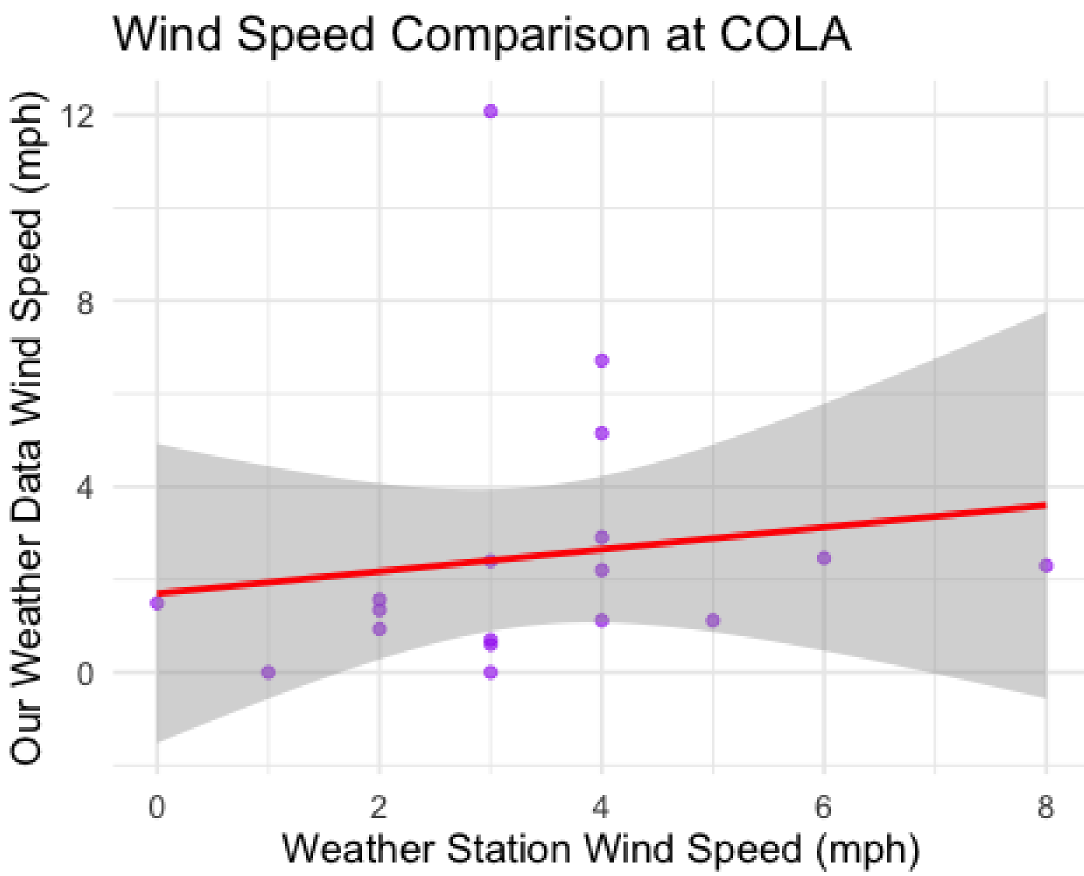

```
## 
## --- Humidity Regression ---
## 
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
## 
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -12.659  -3.870   1.602   4.464   7.476 
## 
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)    
## (Intercept)          5.4731     4.9765   1.100    0.288    
## `Humidity (%)_db`   0.8722     0.1011   8.628 2.05e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 6.13 on 16 degrees of freedom
```

```
## Multiple R-squared:  0.8231, Adjusted R-squared:  0.812
## F-statistic: 74.44 on 1 and 16 DF,  p-value: 2.052e-07

## `geom_smooth()` using formula = 'y ~ x'
```



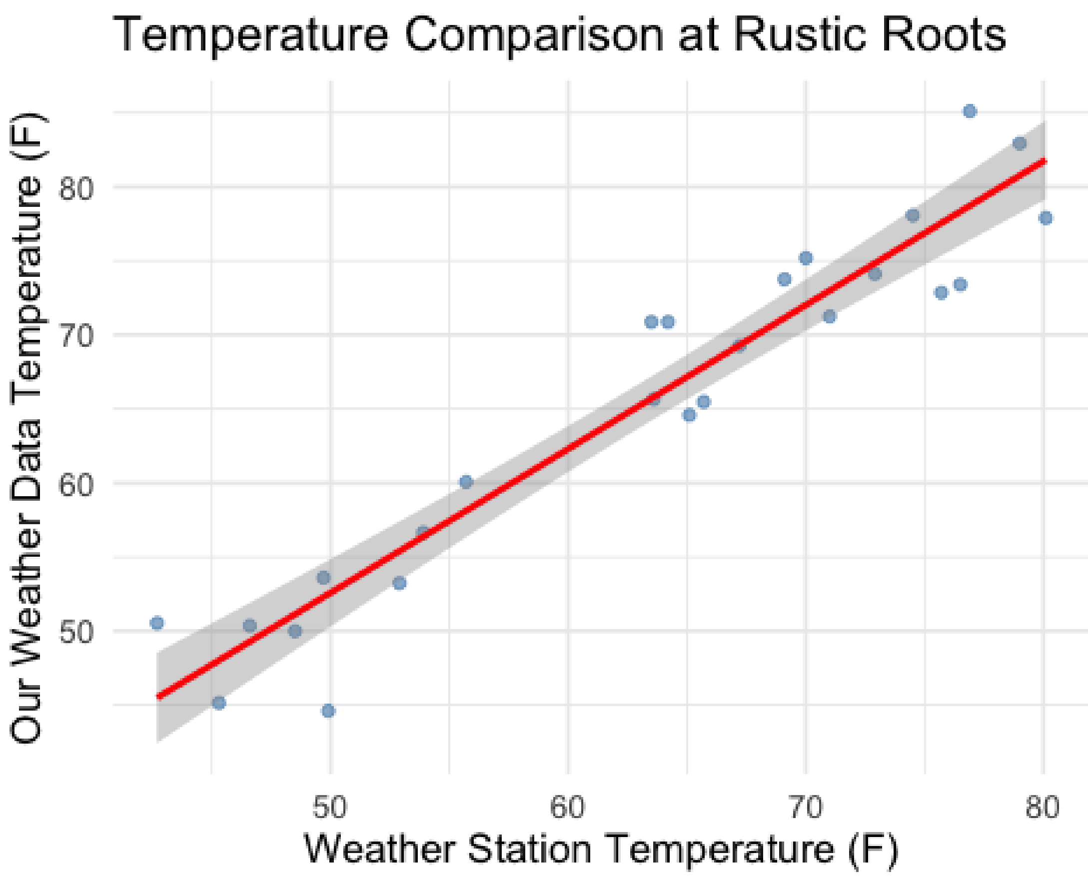

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4117 -1.6664 -0.7477 -0.0608  9.6678
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.6996     1.5205   1.118    0.280
## `WindSpeed (mph)_db`   0.2374     0.3978   0.597    0.559
##
## Residual standard error: 2.984 on 16 degrees of freedom
## Multiple R-squared:  0.02177,    Adjusted R-squared:  -0.03937
## F-statistic: 0.3561 on 1 and 16 DF,  p-value: 0.559
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



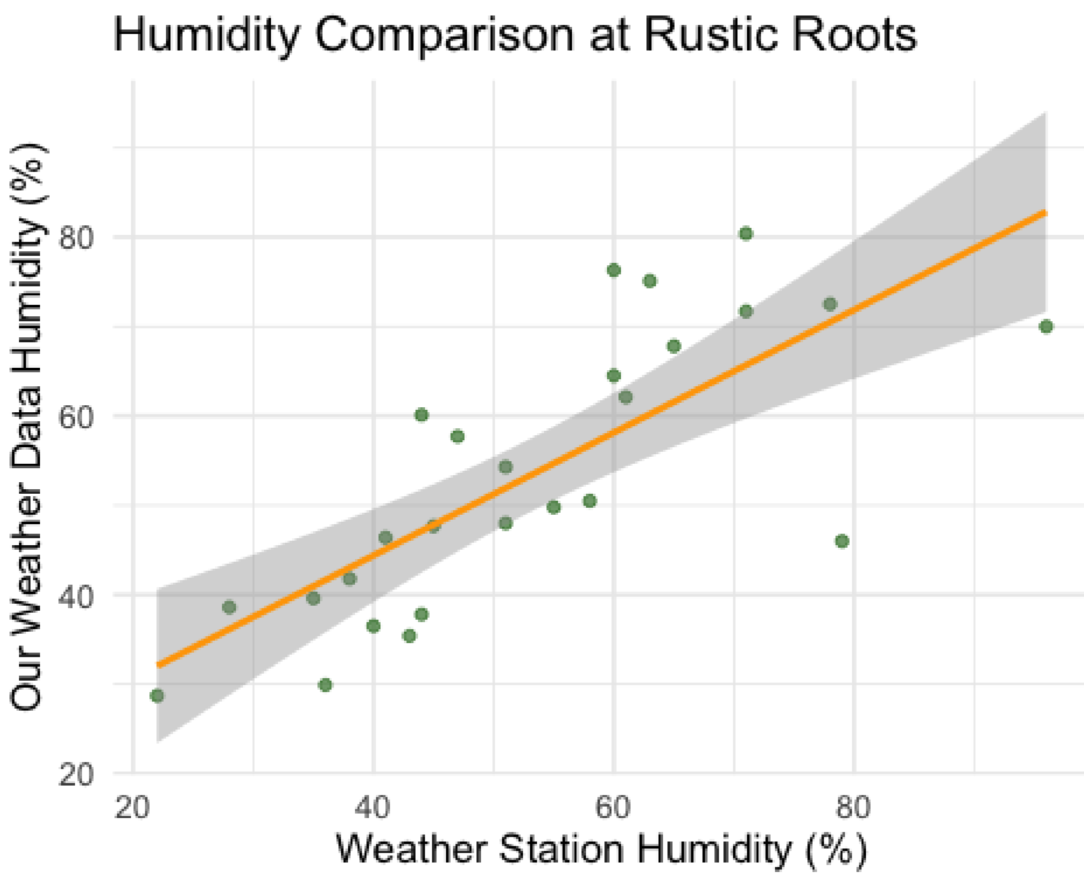

```
##
## ====================
## Location: Rustic Roots
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.8902 -2.3640 -0.0417  2.1716  6.3723
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           3.99940    4.04104    0.99    0.333
## `Temperature (F)_db`  0.97176    0.06291   15.45 1.23e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 3.607 on 23 degrees of freedom
## Multiple R-squared:  0.9121, Adjusted R-squared:  0.9083
## F-statistic: 238.6 on 1 and 23 DF,  p-value: 1.233e-13

## `geom_smooth()` using formula = 'y ~ x'
```



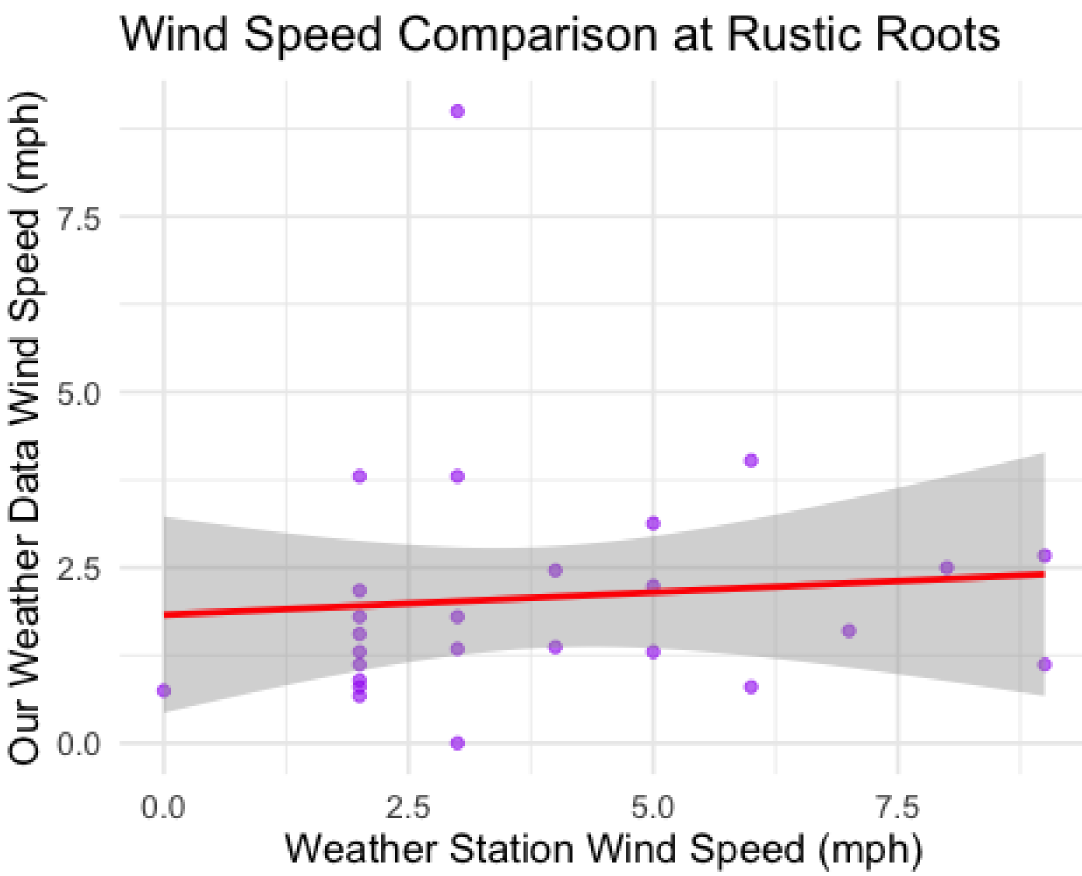

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -25.180  -5.919   0.592   6.179  18.168
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        16.9291     6.5812   2.572   0.0167 *
## `Humidity (%)_db`   0.6867     0.1181   5.817 5.35e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 10.11 on 24 degrees of freedom
## Multiple R-squared:  0.585,  Adjusted R-squared:  0.5677
## F-statistic: 33.84 on 1 and 24 DF,  p-value: 5.355e-06

## `geom_smooth()` using formula = 'y ~ x'
```



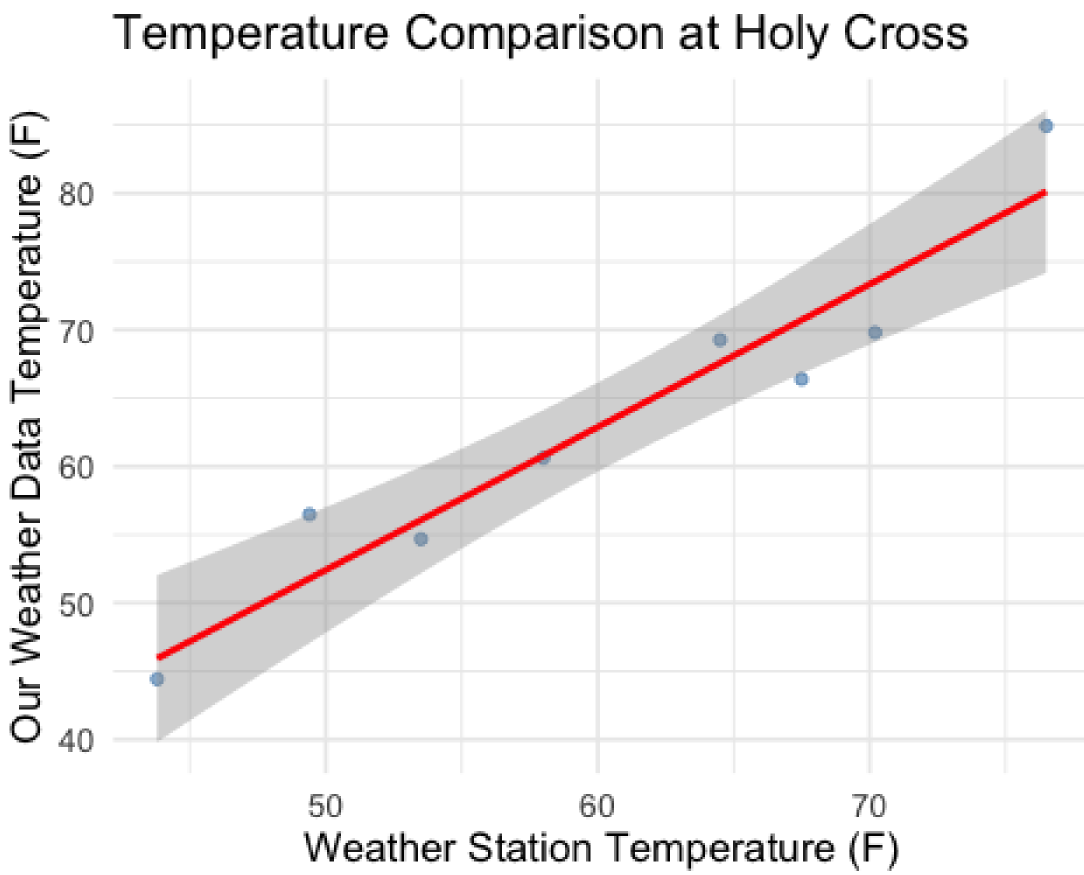

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.0209 -1.0088 -0.5301  0.2541  6.9791
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.82855    0.67759   2.699   0.0125 *
## `WindSpeed (mph)_db`  0.06413    0.14938   0.429   0.6715
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 1.784 on 24 degrees of freedom
## Multiple R-squared:  0.007621,	Adjusted R-squared:  -0.03373
## F-statistic: 0.1843 on 1 and 24 DF,  p-value: 0.6715

## `geom_smooth()` using formula = 'y ~ x'
```



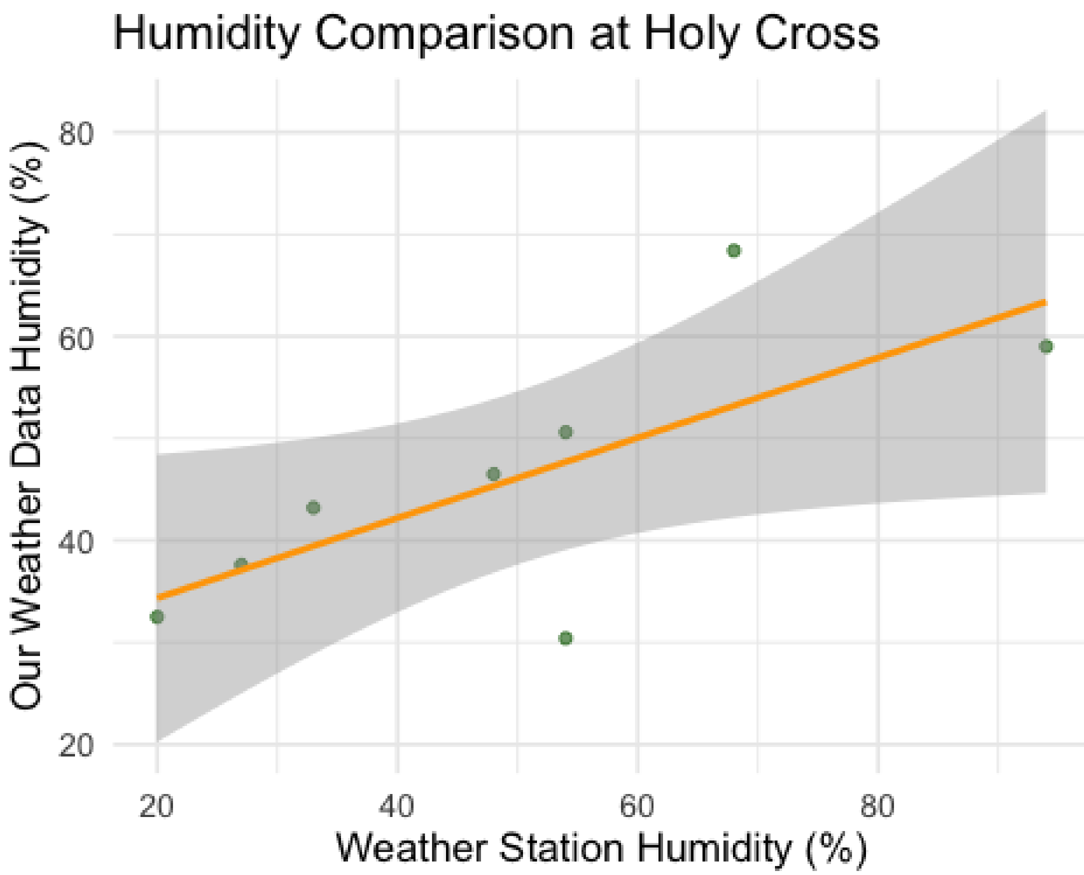

```
##
## ====================
## Location: Holy Cross
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.3396 -2.0701 -0.7805  2.4312  4.7875
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            0.1227     7.8074   0.016 0.987967
```

```
## `Temperature (F)_db`   1.0459     0.1273   8.215 0.000176 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.761 on 6 degrees of freedom
## Multiple R-squared:  0.9183, Adjusted R-squared:  0.9047
## F-statistic: 67.48 on 1 and 6 DF,  p-value: 0.0001756

## `geom_smooth()` using formula = 'y ~ x'
```



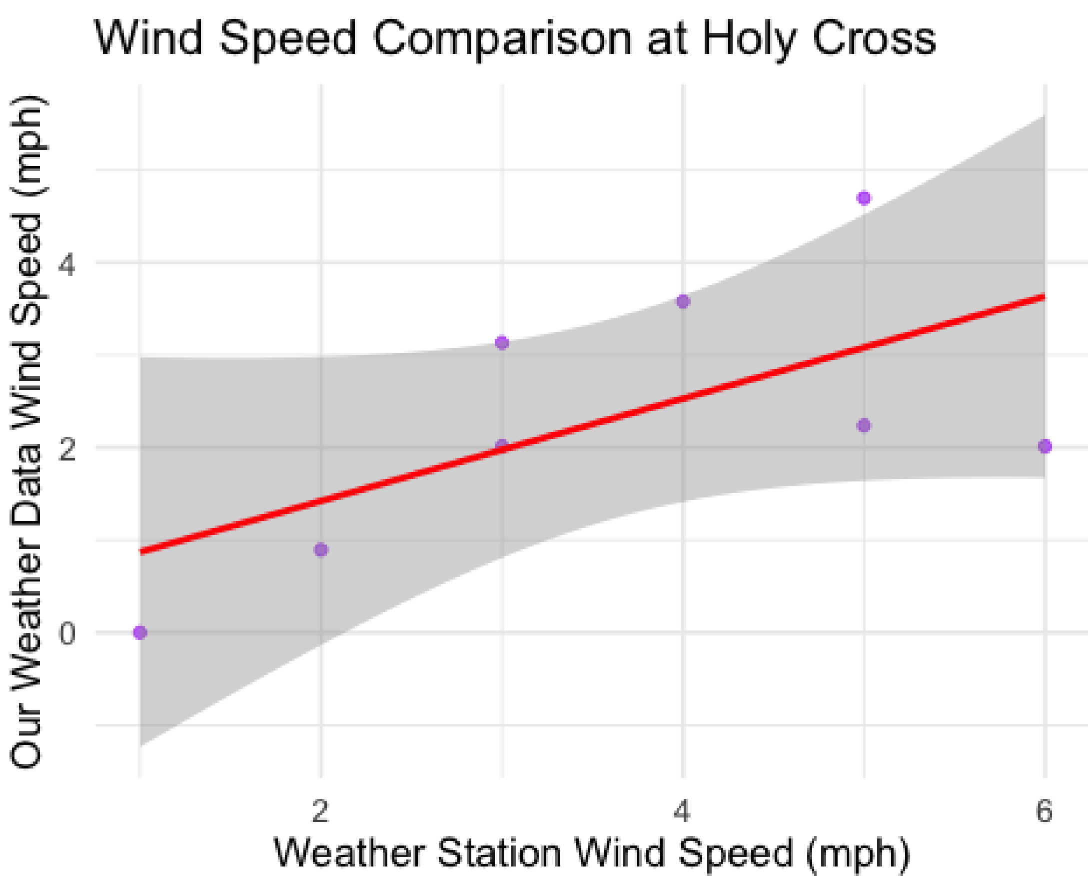

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.2939  -2.4823   0.8354   3.1177  15.2085
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          26.4888     8.4475   3.136   0.0202 *
```

```
## `Humidity (%)_db`   0.3927     0.1548   2.536   0.0443 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.809 on 6 degrees of freedom
## Multiple R-squared:  0.5174, Adjusted R-squared:  0.437
## F-statistic: 6.433 on 1 and 6 DF,  p-value: 0.0443

## `geom_smooth()` using formula = 'y ~ x'
```



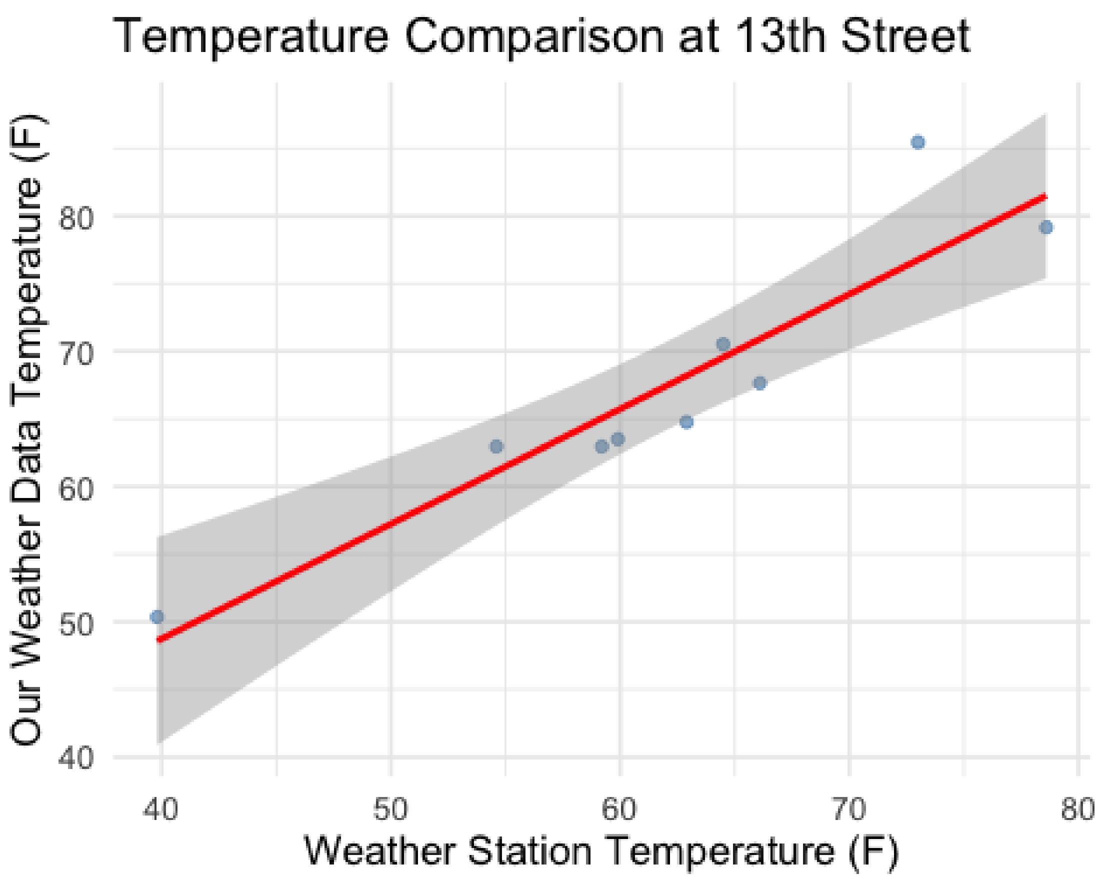

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6207 -0.8505 -0.2448  1.0773  1.6165
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            0.3165     1.1116   0.285   0.7854
```

```
## `WindSpeed (mph)_db`   0.5529     0.2812   1.966   0.0969 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.254 on 6 degrees of freedom
## Multiple R-squared:  0.3918, Adjusted R-squared:  0.2905
## F-statistic: 3.866 on 1 and 6 DF,  p-value: 0.09686

## `geom_smooth()` using formula = 'y ~ x'
```



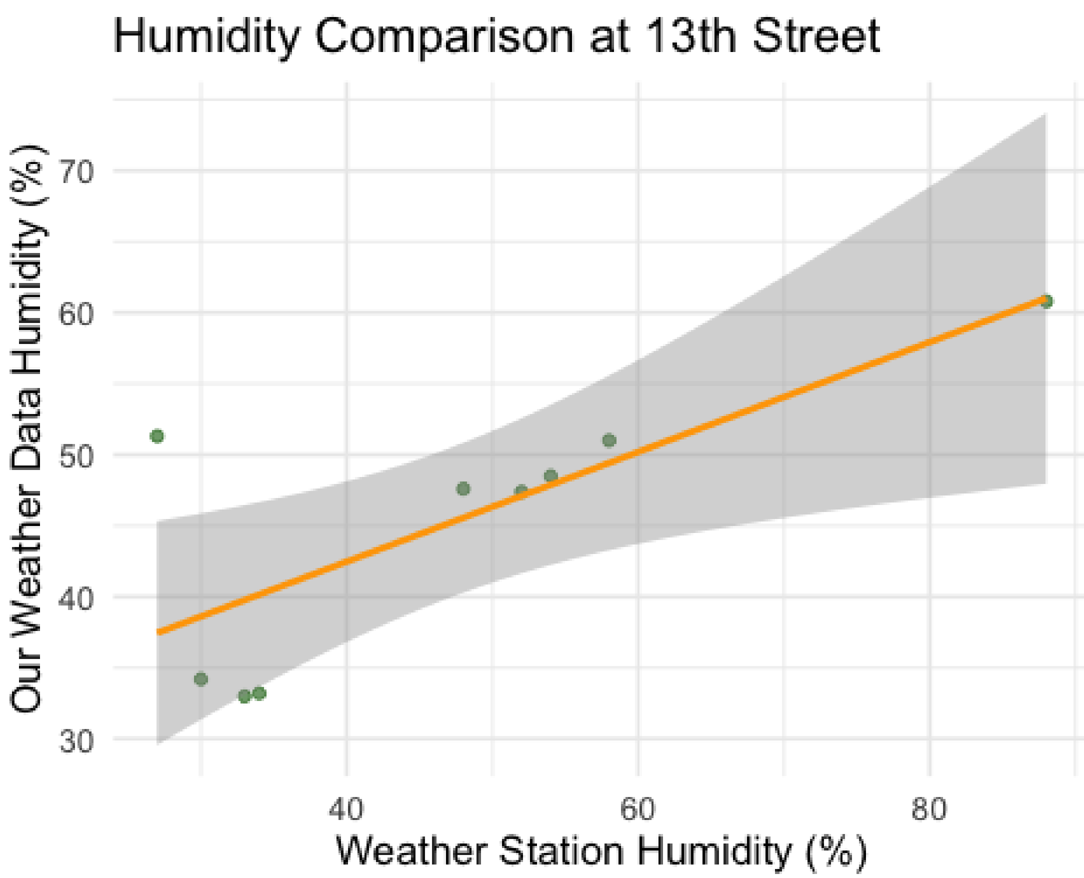

```
##
## ====================
## Location: 13th Street
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.427 -2.353 -2.087  1.779  8.700
```

```
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)    
## (Intercept)            14.7994     8.3226   1.778 0.118601    
## `Temperature (F)_db`    0.8488     0.1322   6.419 0.000361 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.14 on 7 degrees of freedom
## Multiple R-squared:  0.8548, Adjusted R-squared:  0.834 
## F-statistic:  41.2 on 1 and 7 DF,  p-value: 0.0003608

## `geom_smooth()` using formula = 'y ~ x'
```



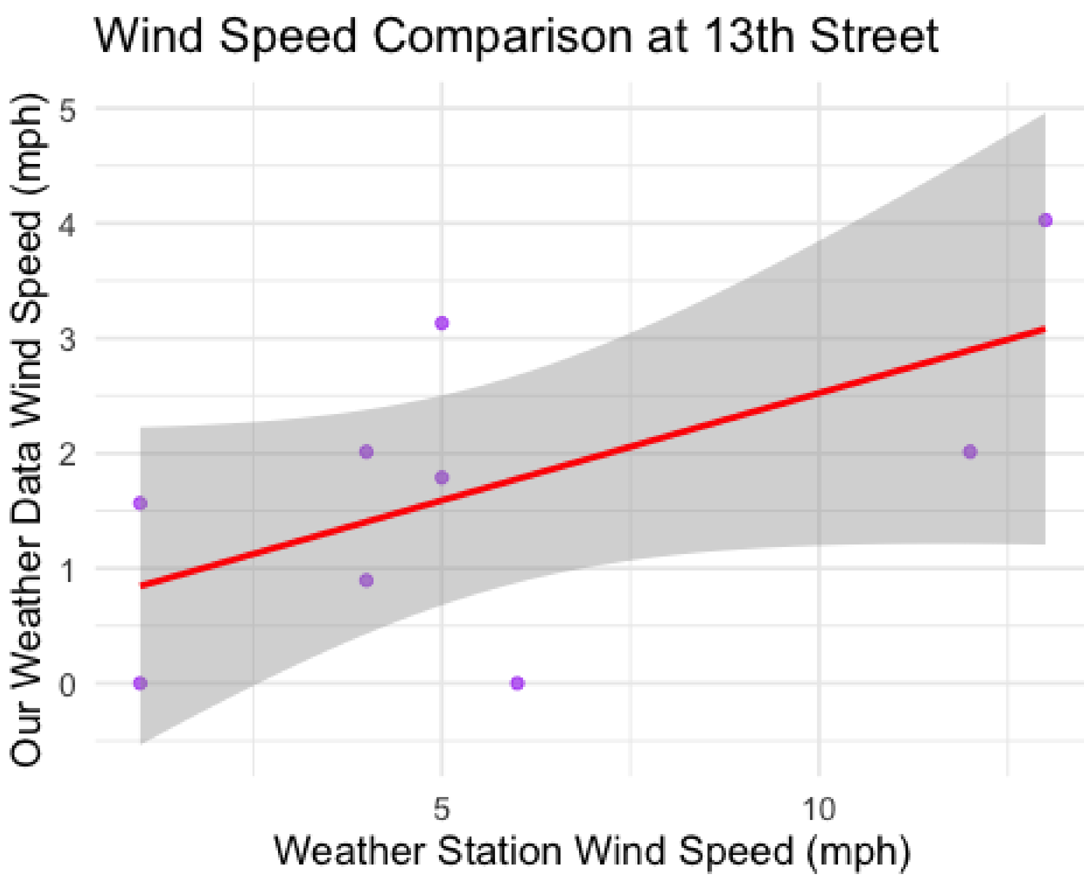

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -6.9597 -4.4153  0.2901  1.5733 13.8431
```

```
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           27.0316     6.2270   4.341  0.00339 **
## `Humidity (%)_db`      0.3861     0.1234   3.128  0.01665 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.682 on 7 degrees of freedom
## Multiple R-squared:  0.583,  Adjusted R-squared:  0.5234
## F-statistic: 9.786 on 1 and 7 DF,  p-value: 0.01665

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7771 -0.8451  0.1988  0.7208  1.5410
```

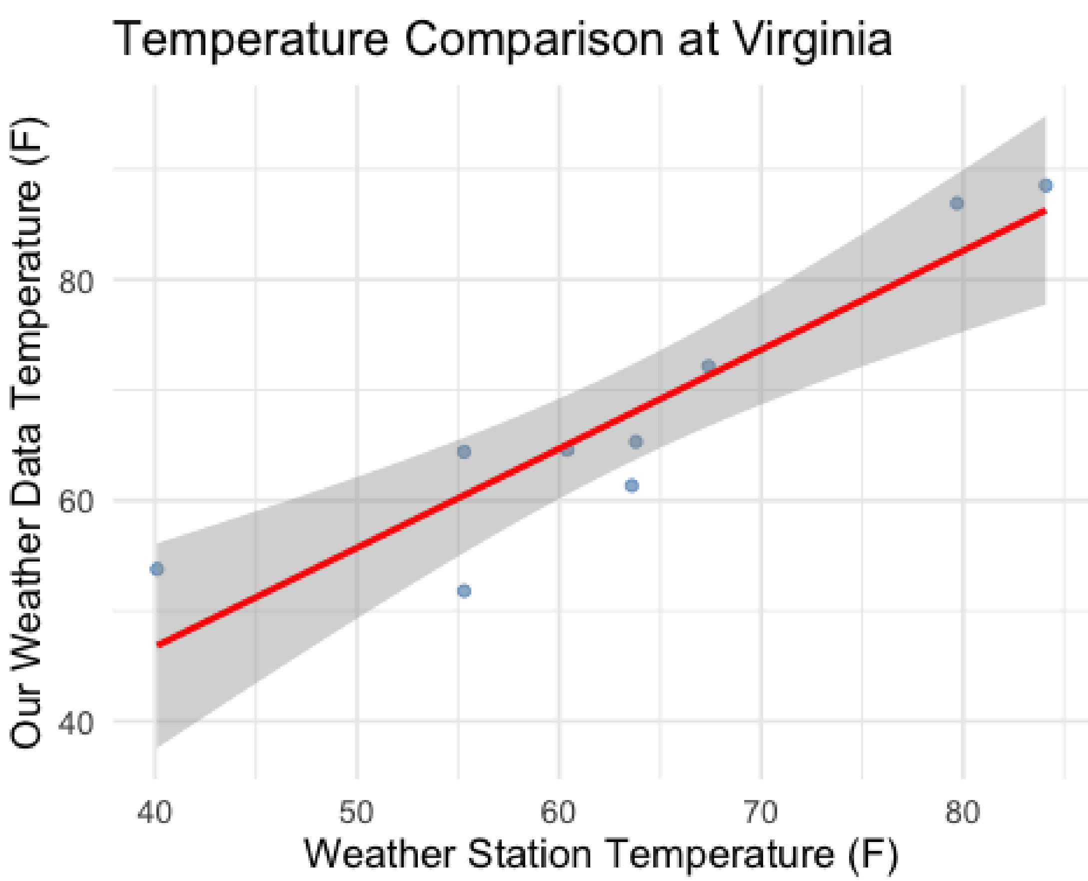
```
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           0.65865    0.65930   0.999   0.3511
## `WindSpeed (mph)_db`  0.18641    0.09505   1.961   0.0907 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.141 on 7 degrees of freedom
## Multiple R-squared:  0.3546, Adjusted R-squared:  0.2624
## F-statistic: 3.846 on 1 and 7 DF,  p-value: 0.09066

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## ====================
## Location: Virginia
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
```

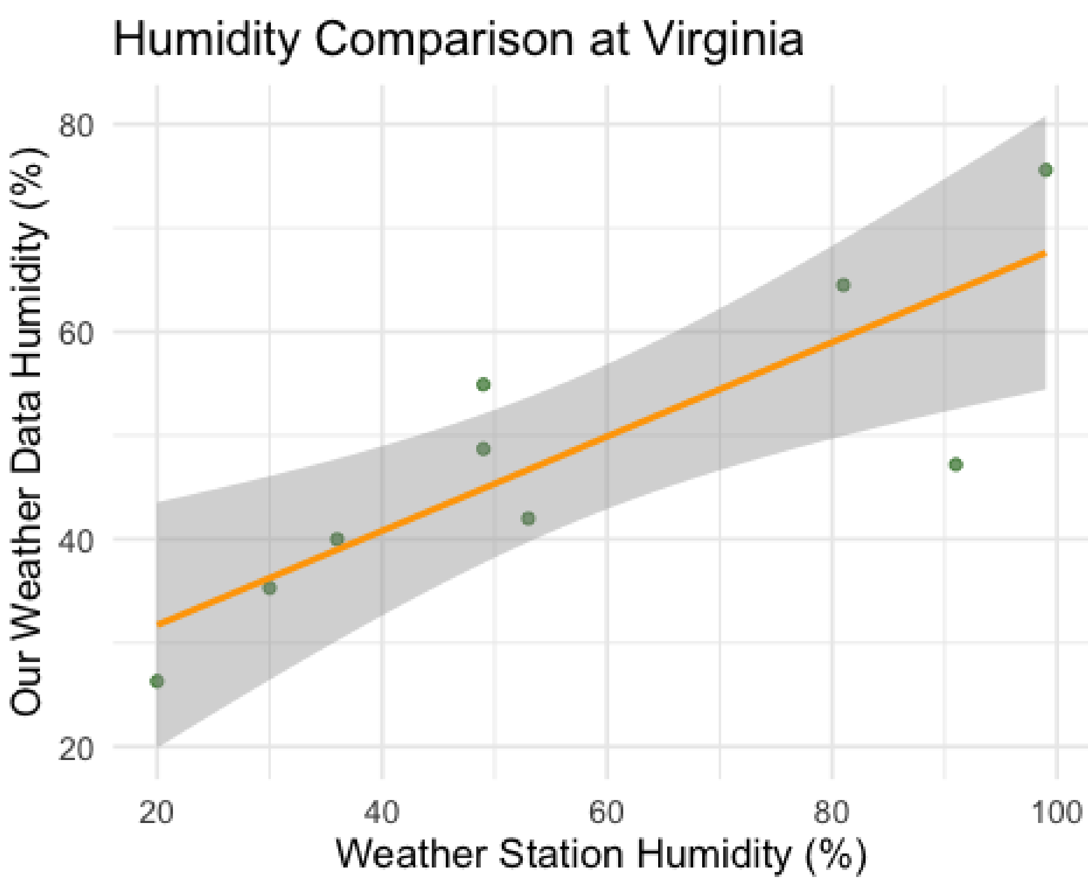
```
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.671 -2.788  0.826  3.929  6.929
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            10.9169     9.6071   1.136 0.293213
## `Temperature (F)_db`    0.8961     0.1489   6.017 0.000533 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.559 on 7 degrees of freedom
## Multiple R-squared:  0.838,  Adjusted R-squared:  0.8148
## F-statistic: 36.21 on 1 and 7 DF,  p-value: 0.000533
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



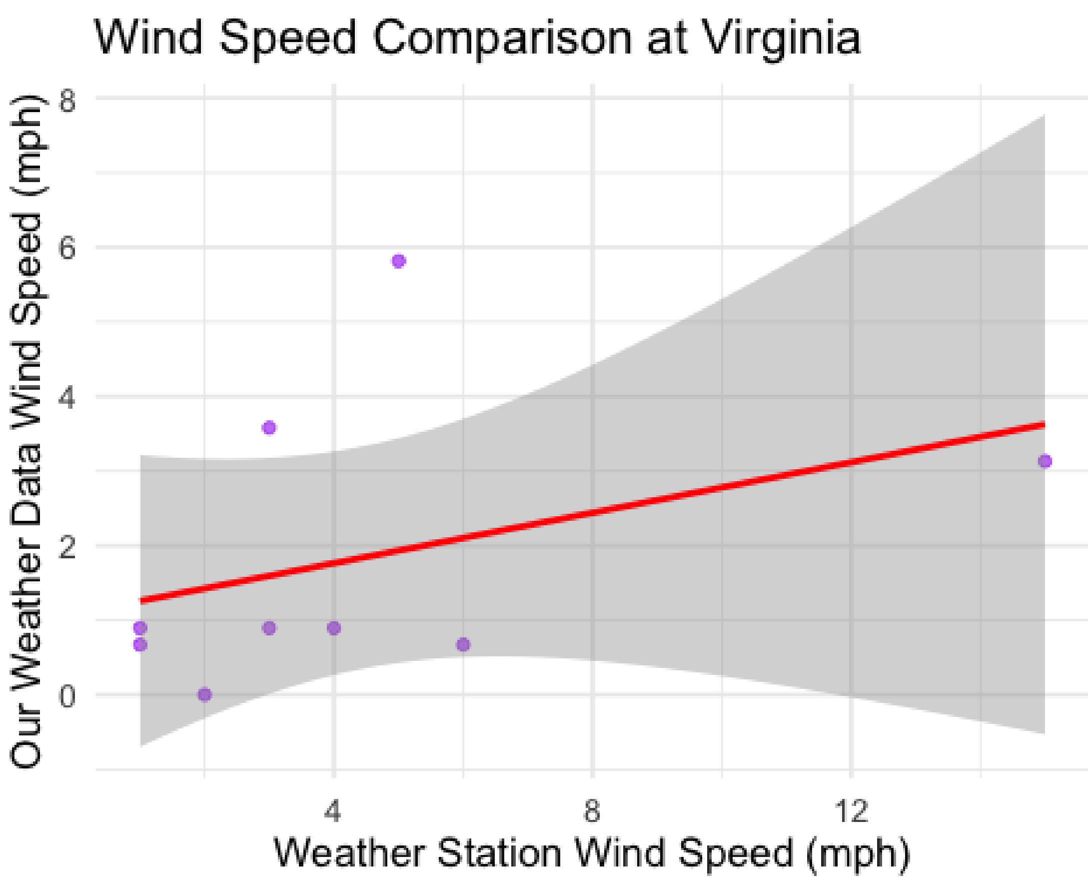

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.779  -4.713   1.012   5.064  10.005
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          22.6302     6.9484   3.257  0.01392 *
## `Humidity (%)_db`     0.4544     0.1117   4.069  0.00475 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.775 on 7 degrees of freedom
## Multiple R-squared:  0.7029, Adjusted R-squared:  0.6604
## F-statistic: 16.56 on 1 and 7 DF,  p-value: 0.004753

## `geom_smooth()` using formula = 'y ~ x'
```



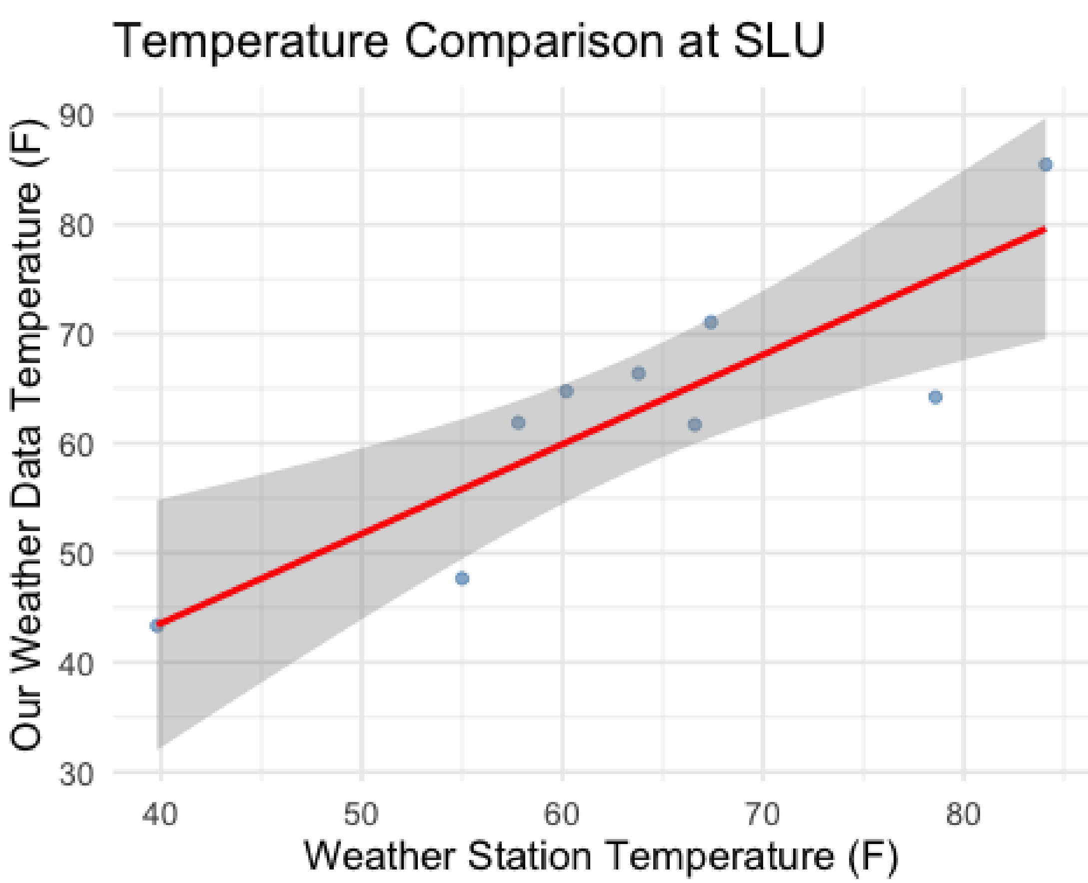

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.4314 -0.8693 -0.5854 -0.3617  3.8828
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.0873     0.9353   1.162    0.283
## `WindSpeed (mph)_db`   0.1692     0.1554   1.089    0.312
##
## Residual standard error: 1.892 on 7 degrees of freedom
## Multiple R-squared:  0.1448, Adjusted R-squared:  0.02265
## F-statistic: 1.185 on 1 and 7 DF,  p-value: 0.3123

## `geom_smooth()` using formula = 'y ~ x'
```



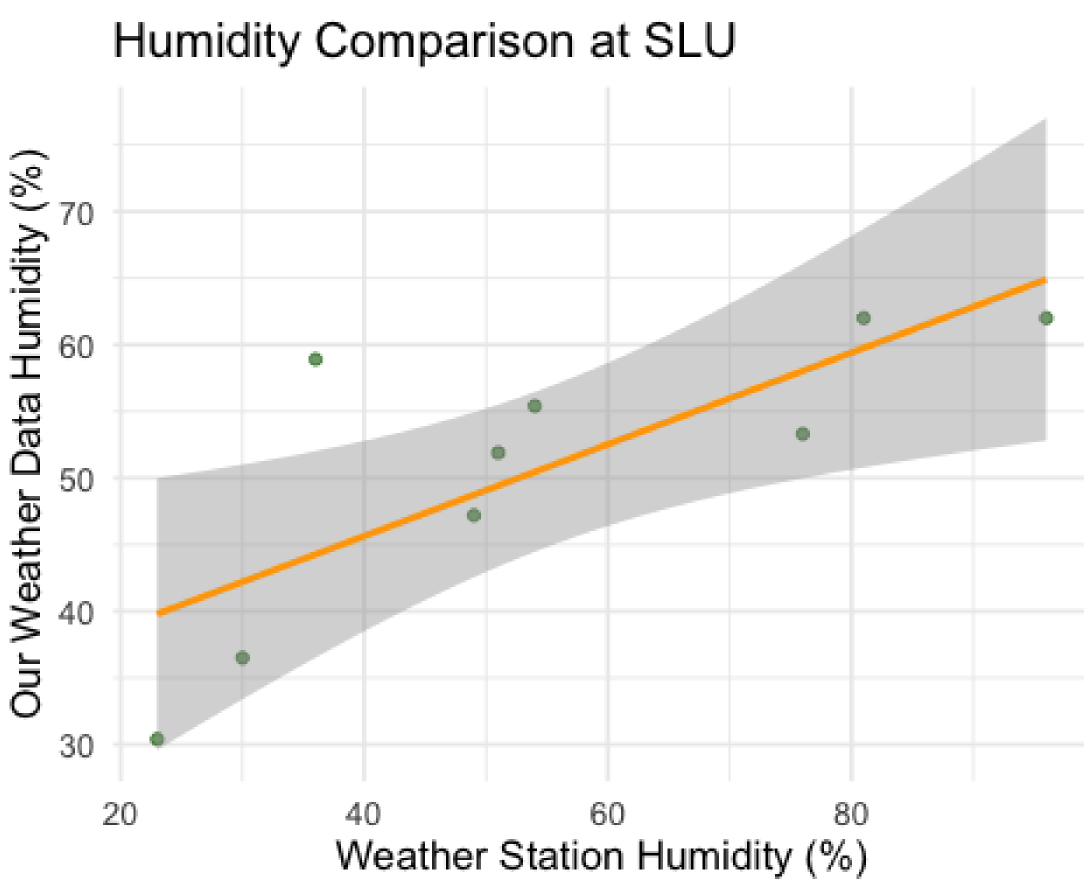

```
##
## ====================
## Location: SLU
## ====================
##
## --- Temperature Regression ---
##
```

```
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.895  -3.610   3.358   4.680   5.851
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           10.8916    11.6556   0.934  0.38118
## `Temperature (F)_db`   0.8171     0.1797   4.548  0.00264 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.612 on 7 degrees of freedom
## Multiple R-squared:  0.7471, Adjusted R-squared:  0.711
## F-statistic: 20.68 on 1 and 7 DF,  p-value: 0.002644

## `geom_smooth()` using formula = 'y ~ x'
```



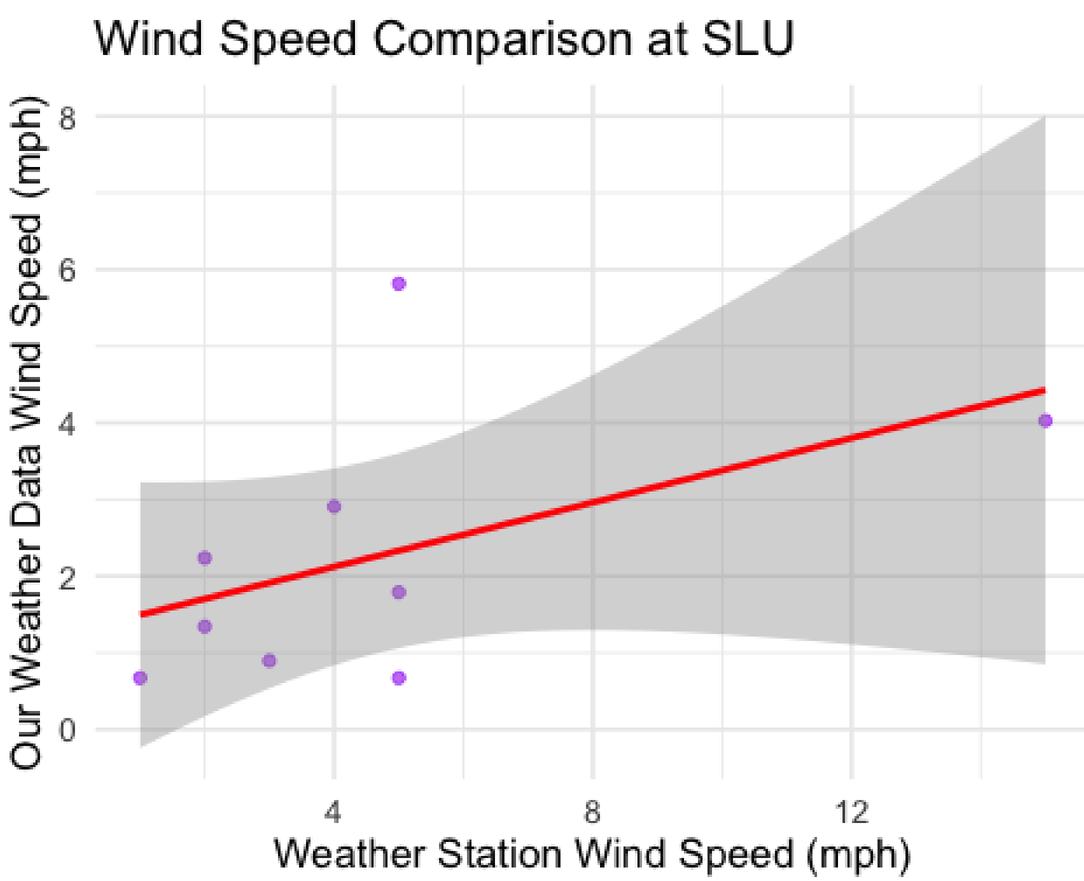

```
##
## --- Humidity Regression ---
##
```

```
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -9.400 -4.729 -1.543  2.470 14.629
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        31.8895     6.5147   4.895  0.00176 **
## `Humidity (%)_db`   0.3439     0.1089   3.158  0.01598 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.594 on 7 degrees of freedom
## Multiple R-squared:  0.5875, Adjusted R-squared:  0.5286
## F-statistic: 9.971 on 1 and 7 DF,  p-value: 0.01598

## `geom_smooth()` using formula = 'y ~ x'
```



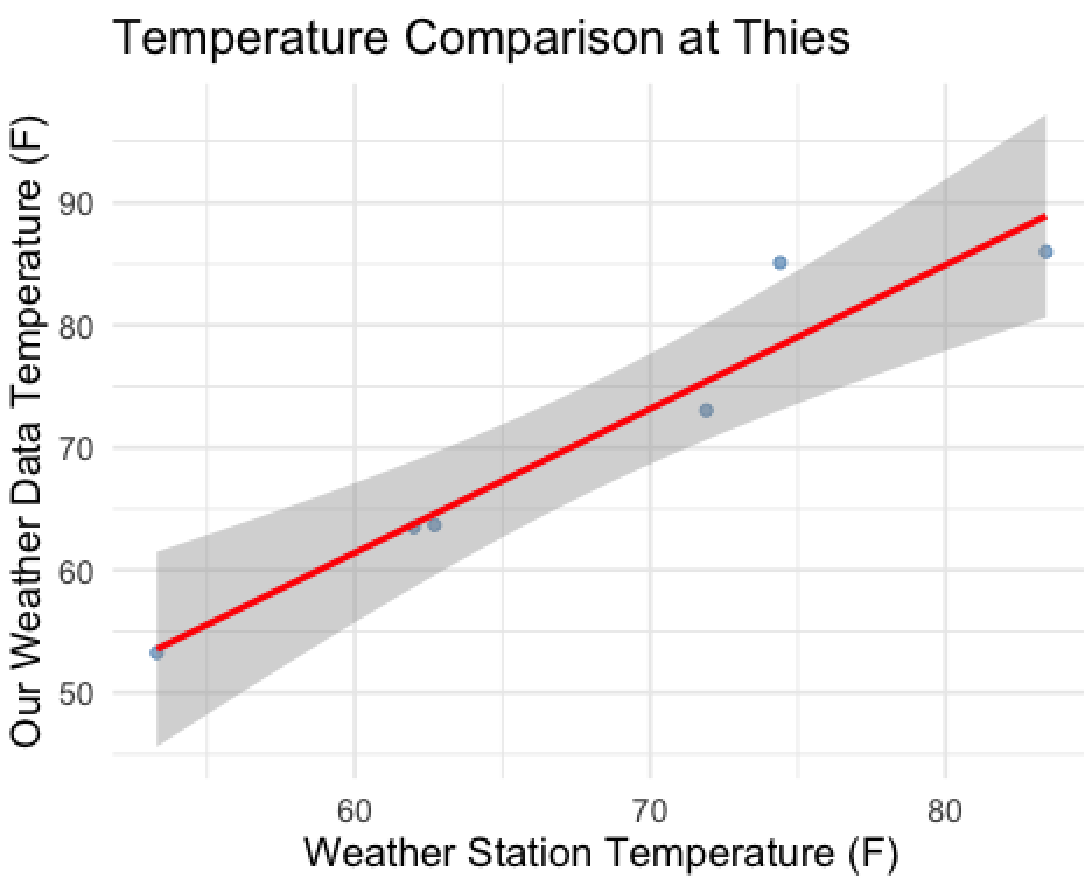

```
##
## --- Wind Speed Regression ---
##
```

```
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6606 -0.8220 -0.4016  0.5342  3.4844
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.2834     0.8338   1.539    0.168
## `WindSpeed (mph)_db`   0.2096     0.1369   1.532    0.169
##
## Residual standard error: 1.608 on 7 degrees of freedom
## Multiple R-squared:  0.251,  Adjusted R-squared:  0.144
## F-statistic: 2.346 on 1 and 7 DF,  p-value: 0.1695

## `geom_smooth()` using formula = 'y ~ x'
```

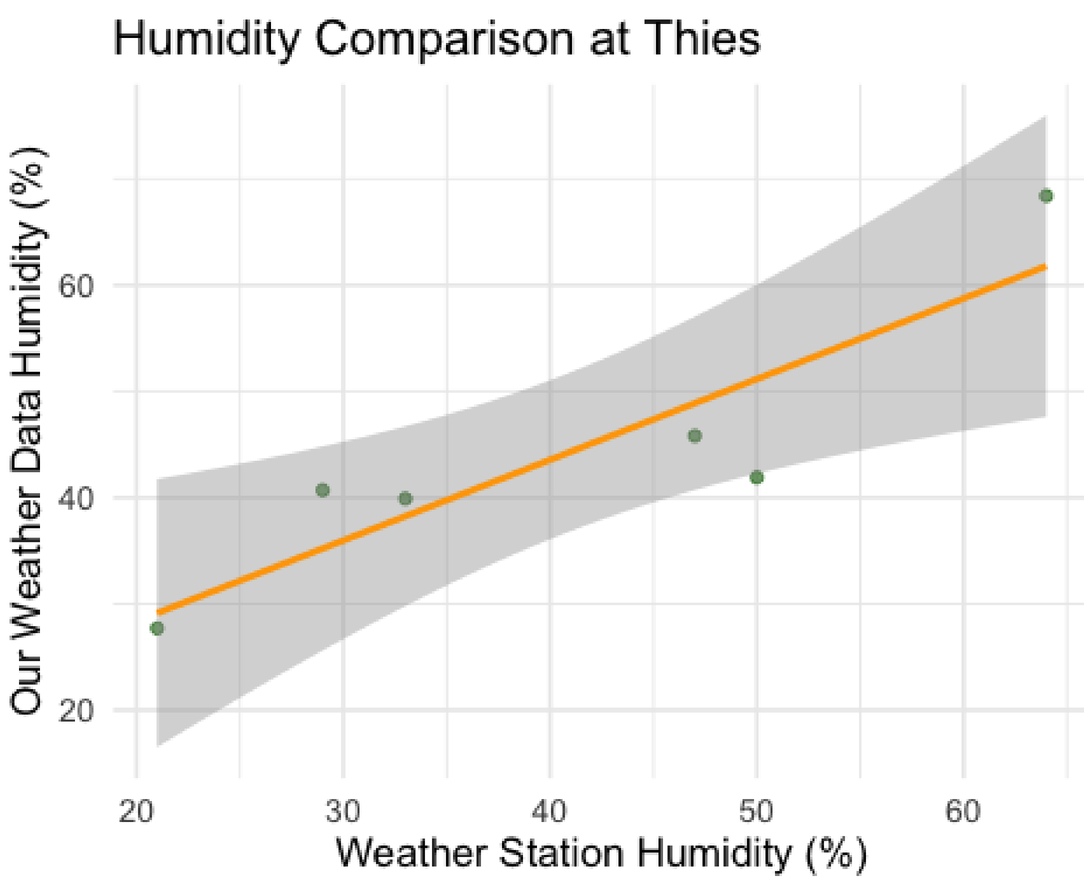



```
##
## ====================
## Location: Thies
## ====================
##
```

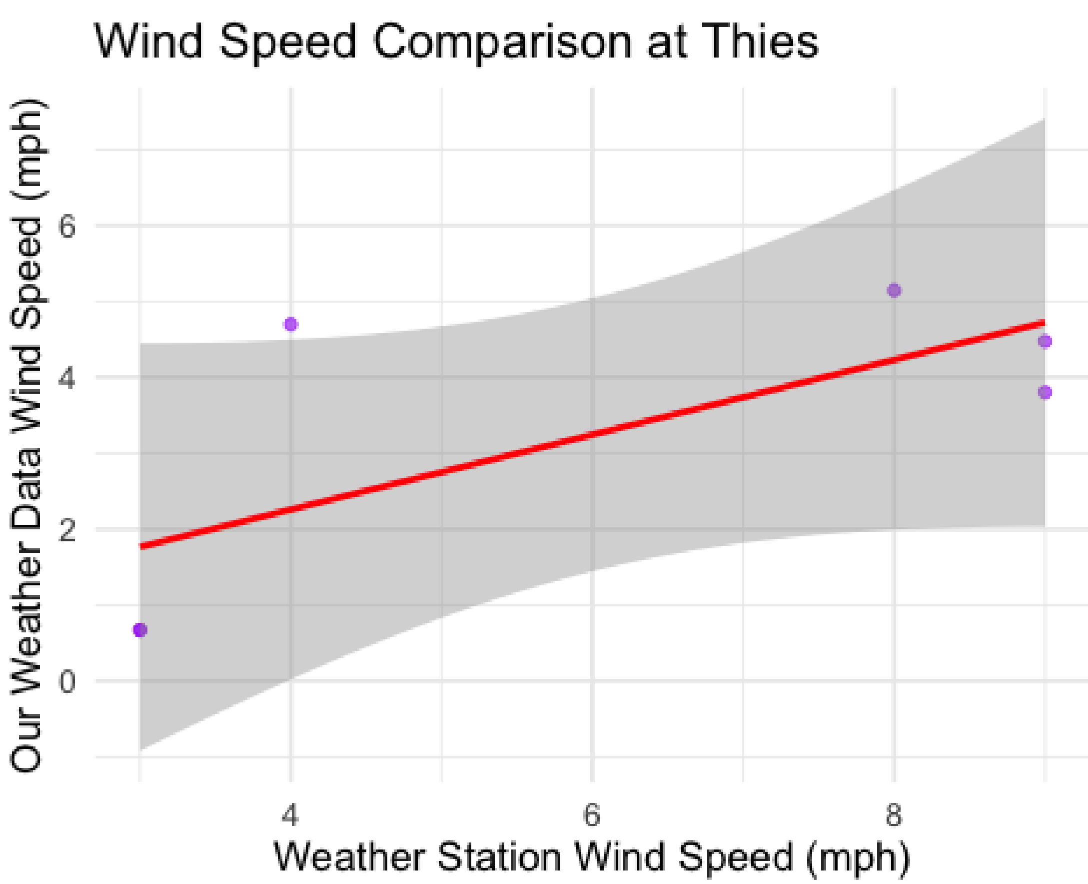
```
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##       1       2       3       4       5       6
## -0.2673  6.7597 -0.9100 -2.9174 -0.3028 -2.3622
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           -9.0973    11.1741  -0.814  0.46125
## `Temperature (F)_db`   1.1752     0.1628   7.220  0.00195 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.898 on 4 degrees of freedom
## Multiple R-squared:  0.9287, Adjusted R-squared:  0.9109
## F-statistic: 52.13 on 1 and 4 DF,  p-value: 0.001951

## `geom_smooth()` using formula = 'y ~ x'
```



```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##      1      2      3      4      5      6
## -1.434 -9.253 -3.075  1.654  5.492  6.617
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        13.1892     8.0362   1.641   0.1761
## `Humidity (%)_db`   0.7593     0.1862   4.078   0.0151 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.592 on 4 degrees of freedom
## Multiple R-squared:  0.8061, Adjusted R-squared:  0.7576
## F-statistic: 16.63 on 1 and 4 DF,  p-value: 0.01513

## `geom_smooth()` using formula = 'y ~ x'
```

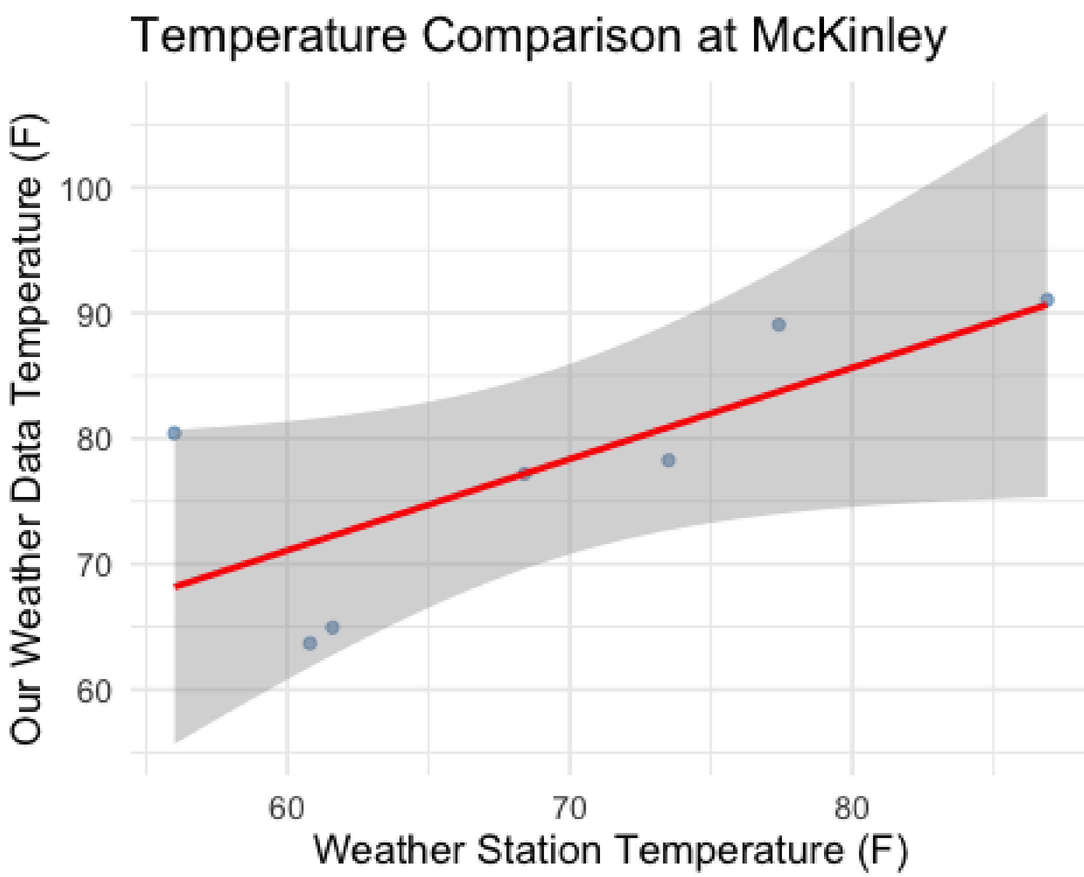


```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##       1       2       3       4       5       6
## -0.9202 -1.0931 -0.2491 -1.0931  2.4403  0.9151
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)             0.2847     1.5762   0.181    0.865
## `WindSpeed (mph)_db`    0.4931     0.2394   2.060    0.109
##
## Residual standard error: 1.588 on 4 degrees of freedom
## Multiple R-squared:  0.5147, Adjusted R-squared:  0.3933
## F-statistic: 4.242 on 1 and 4 DF,  p-value: 0.1085

## `geom_smooth()` using formula = 'y ~ x'
```

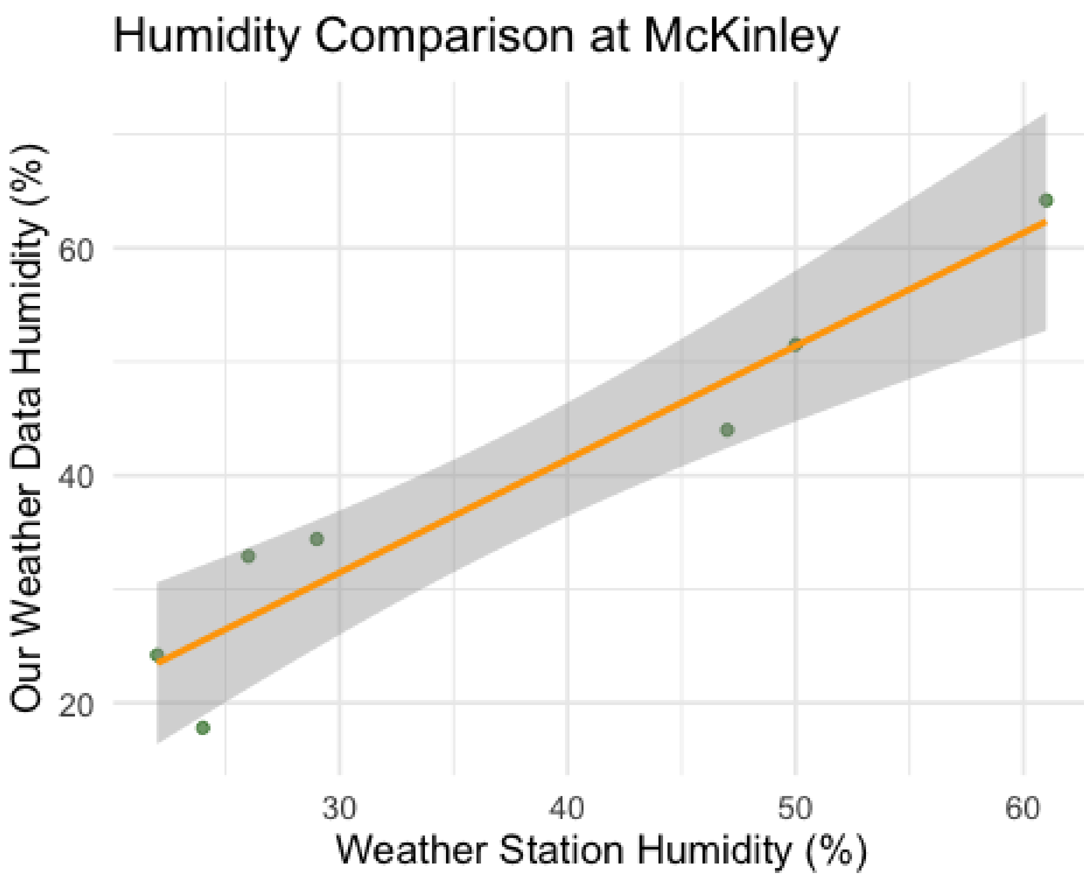



```
##
## ====================
```

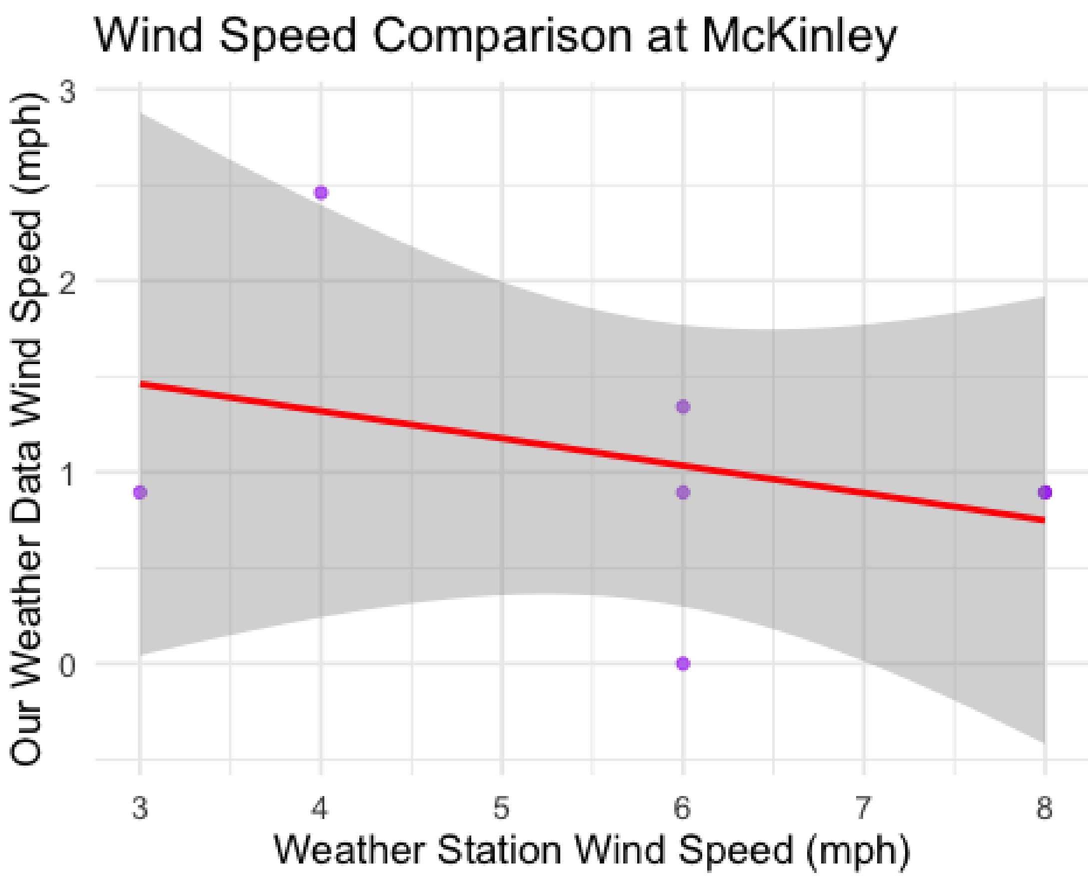
```
## Location: McKinley
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
##        1        2        3        4        5        6        7
## -7.30677  5.31752 -0.01429  0.38554 12.24765 -2.64493 -7.98471
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)           27.4281    20.5342   1.336   0.2392
## `Temperature (F)_db`   0.7276     0.2936   2.478   0.0559 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.779 on 5 degrees of freedom
## Multiple R-squared:  0.5513, Adjusted R-squared:  0.4615
## F-statistic: 6.143 on 1 and 5 DF,  p-value: 0.05595

## `geom_smooth()` using formula = 'y ~ x'
```



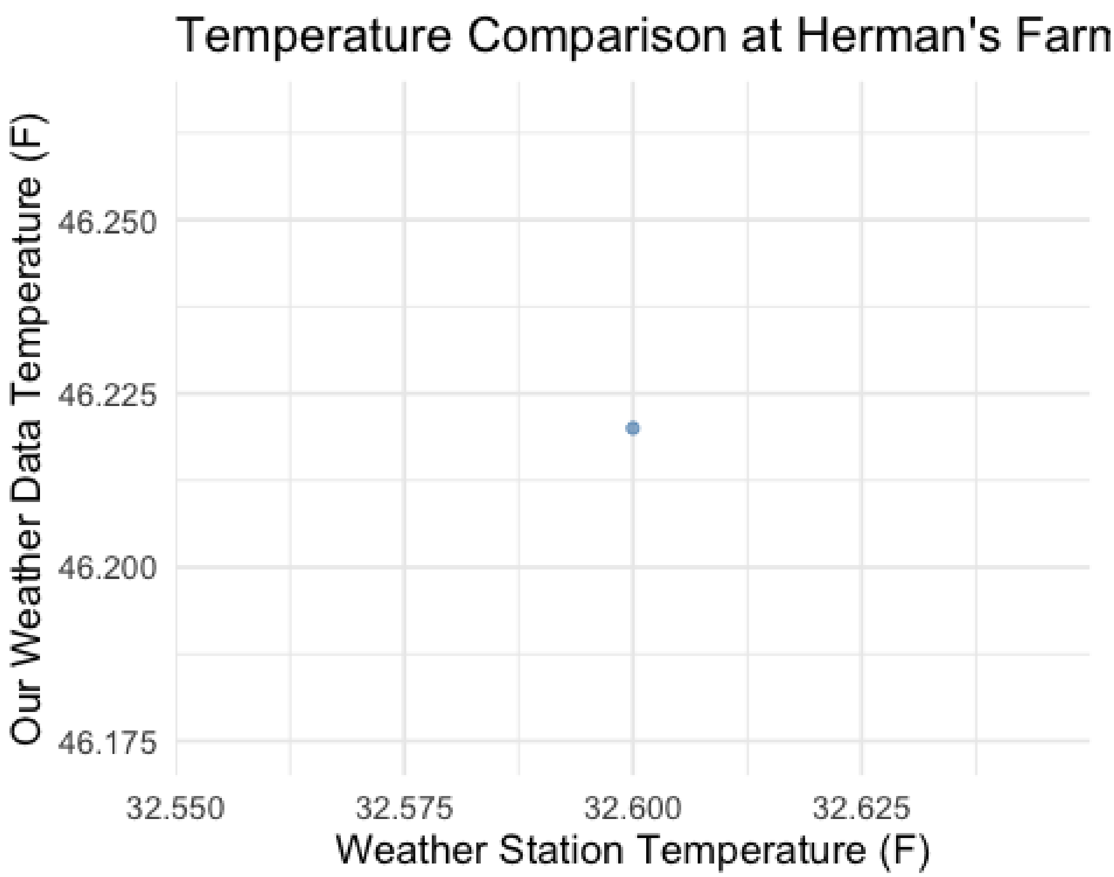

```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
##       1       2       3       4       5       6       7
##  0.7167 -4.3921  0.1189  3.9422 -7.6760  1.8591  5.4313
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1.5637     5.3009   0.295 0.779861
## `Humidity (%)_db`  0.9963     0.1337   7.453 0.000686 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.045 on 5 degrees of freedom
## Multiple R-squared:  0.9174, Adjusted R-squared:  0.9009
## F-statistic: 55.55 on 1 and 5 DF,  p-value: 0.0006859

## `geom_smooth()` using formula = 'y ~ x'
```



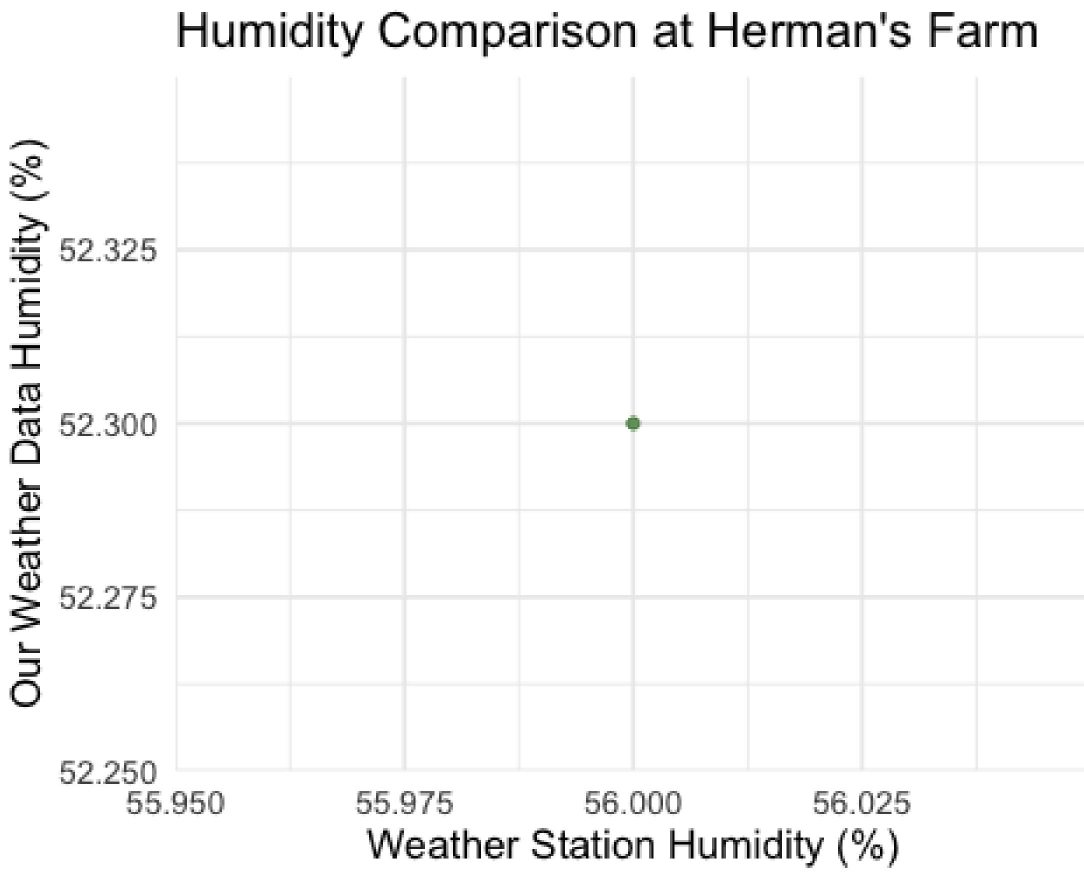

```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
##       1       2       3       4       5       6       7
##  0.3080  0.1456  0.1456 -0.5669 -1.0342 -0.1394  1.1415
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)            1.8891     1.0095   1.871    0.120
## `WindSpeed (mph)_db`  -0.1425     0.1653  -0.862    0.428
##
## Residual standard error: 0.7551 on 5 degrees of freedom
## Multiple R-squared:  0.1293, Adjusted R-squared:  -0.04479
## F-statistic: 0.7428 on 1 and 5 DF,  p-value: 0.4282

## `geom_smooth()` using formula = 'y ~ x'
```



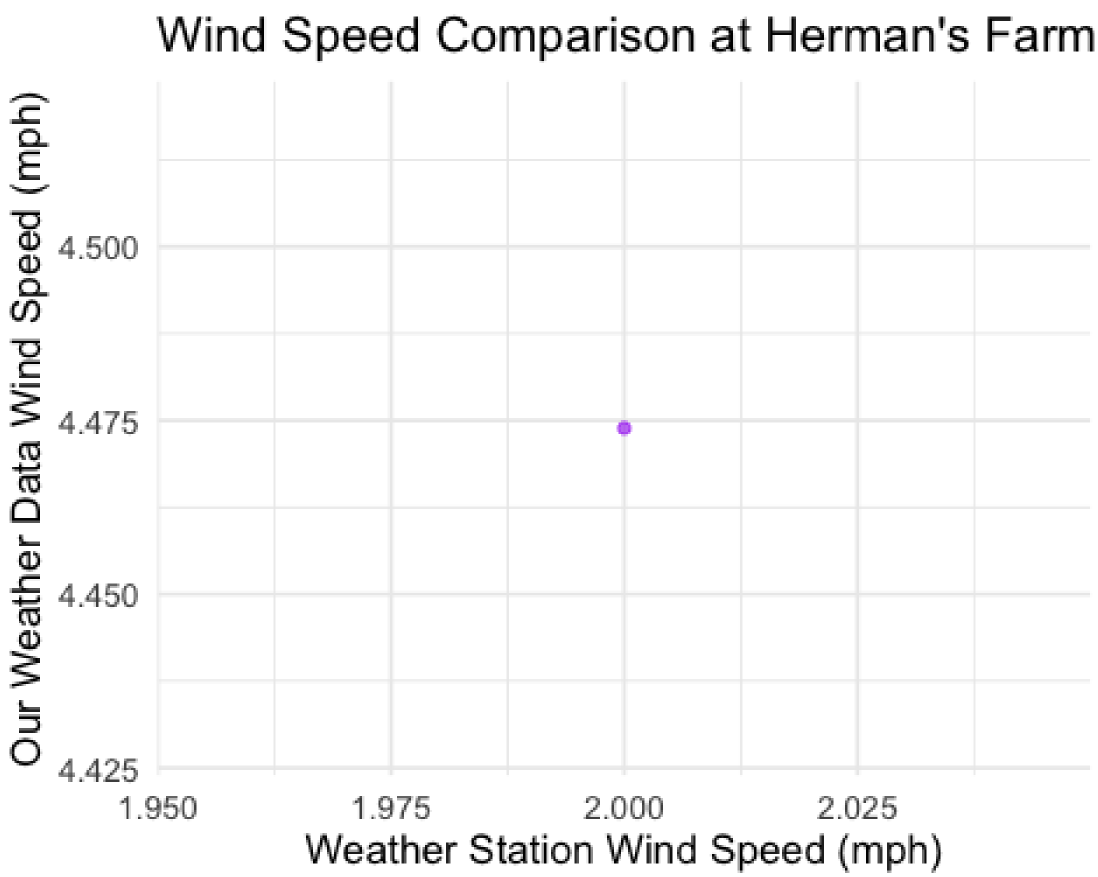

```
##
## ====================
## Location: Herman's Farm
## ====================
##
## --- Temperature Regression ---
##
## Call:
## lm(formula = `Temperature (F)` ~ `Temperature (F)_db`, data = temp_data)
##
## Residuals:
## ALL 1 residuals are 0: no residual degrees of freedom!
##
## Coefficients: (1 not defined because of singularities)
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)             46.22        NaN     NaN      NaN
## `Temperature (F)_db`       NA         NA      NA       NA
##
## Residual standard error: NaN on 0 degrees of freedom

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Humidity Regression ---
##
## Call:
## lm(formula = `Humidity (%)` ~ `Humidity (%)_db`, data = hum_data)
##
## Residuals:
## ALL 1 residuals are 0: no residual degrees of freedom!
##
## Coefficients: (1 not defined because of singularities)
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)           52.3        NaN     NaN      NaN
## `Humidity (%)_db`       NA         NA      NA       NA
##
## Residual standard error: NaN on 0 degrees of freedom

## `geom_smooth()` using formula = 'y ~ x'
```




```
##
## --- Wind Speed Regression ---
##
## Call:
## lm(formula = `WindSpeed (mph)` ~ `WindSpeed (mph)_db`, data = wind_data)
##
## Residuals:
## ALL 1 residuals are 0: no residual degrees of freedom!
##
## Coefficients: (1 not defined because of singularities)
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)              4.474        NaN     NaN      NaN
## `WindSpeed (mph)_db`        NA         NA      NA       NA
##
## Residual standard error: NaN on 0 degrees of freedom

## `geom_smooth()` using formula = 'y ~ x'
```


Wind Speed Comparison at Herman's Farm
Our Weather Data Wind Speed (mph)
4.500
4.475
4.450
4.425
1.950
1.975
2.000
2.025
Weather Station Wind Speed (mph)